BACHELOR IN COMPUTER SCIENCE (HONOURS)

DEVELOPMENT OF SMART PREDICTIVE MAINTENANCE SYSTEM FOR MANUFACTARING COMPANY

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DEVELOPMENT OF SMART PREDICTIVE MAINTENANCE SYSTEM FOR MANUFACTARING COMPANY

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ABSTRACT

The increasing reliance on machinery in manufacturing has exposed weaknesses in traditional maintenance practices, particularly among small and medium enterprises (SMEs). Reactive and preventive maintenance approaches often result in unexpected equipment failures, extended downtime, and inefficient use of resources. This project addresses these issues by proposing a smart predictive maintenance framework to enhance maintenance efficiency and decision-making. The proposed framework integrates telemetry data acquisition, supervised machine learning techniques, and a user-friendly monitoring dashboard to enable early fault detection and automated maintenance alerts. The system is designed based on the V-Model software development lifecycle to ensure systematic verification and validation at each development stage. Machine learning classifiers, namely Random Forest and Support Vector Machine models, are trained using simulated industrial datasets to classify machine operating conditions and predict potential failures. Real-time data collection and historical data storage are implemented to support performance monitoring and trend analysis. The system is also integrated with previously siloed operational systems to improve maintenance visibility and information sharing. The results demonstrate that the proposed framework can reduce unplanned downtime, improve fault detection accuracy, and support informed maintenance decisions. In conclusion, the proposed predictive maintenance system addresses key Industry 4.0 challenges and supports the digital transformation of manufacturing operations, particularly for small and medium enterprises.

ABSTRAK

Kebergantungan yang semakin meningkat terhadap mesin dalam sektor pembuatan telah mendedahkan kelemahan dalam amalan penyelenggaraan tradisional, khususnya dalam kalangan perusahaan kecil dan sederhana (PKS). Pendekatan penyelenggaraan reaktif dan pencegahan sering menyebabkan kegagalan peralatan yang tidak dijangka, masa henti operasi yang panjang, serta penggunaan sumber yang tidak cekap. Projek ini mencadangkan satu rangka kerja penyelenggaraan ramalan pintar bagi meningkatkan kecekapan penyelenggaraan dan sokongan pembuatan keputusan. Rangka kerja yang dicadangkan menggabungkan pemerolehan data telemetri, teknik pembelajaran mesin terawasi, dan papan pemuka pemantauan mesra pengguna untuk pengesanan awal kerosakan serta penghasilan amaran penyelenggaraan secara automatik. Sistem ini direka berdasarkan kitaran pembangunan perisian V-Model bagi memastikan proses pengesahan dan pengesahan semula dilaksanakan secara sistematik. Model Random Forest dan Support Vector Machine dilatih menggunakan set data industri simulasi untuk mengklasifikasikan keadaan operasi mesin dan meramal kegagalan. Sistem ini menyokong pengumpulan data masa nyata dan penyimpanan data sejarah bagi tujuan pemantauan prestasi dan analisis trend, serta disepadukan dengan sistem operasi berasingan untuk meningkatkan keterlihatan penyelenggaraan. Keputusan menunjukkan bahawa rangka kerja ini berupaya mengurangkan masa henti tidak dirancang dan menyokong keputusan penyelenggaraan yang lebih berkesan. Kesimpulannya, sistem penyelenggaraan ramalan yang dicadangkan menyokong cabaran Industri 4.0 dan transformasi digital dalam sektor pembuatan, khususnya bagi perusahaan kecil dan sederhana.

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CHAPTER 1

INTRODUCTION

1.1 Project Background

In today's manufacturing world, keeping machines running smoothly is absolutely essential to avoid interruptions in production. With competition getting tougher and customer demand constantly changing, manufacturers are under more pressure than ever to cut down on downtime and lower the costs that come with maintenance. Yet, many still rely on traditional methods like fixing machines only after they break or following fixed schedules for preventive maintenance. While these approaches are common, they often miss the mark,machines might break unexpectedly or get serviced more often than needed. This not only wastes valuable resources but also messes with production timelines and eats into profits.

Luckily, recent advances in technology have brought about Predictive Maintenance (PdM) systems. These smart systems use live sensor data combined with machine learning to spot early warning signs and predict when a machine might fail, so problems can be addressed before they happen. Predictive maintenance fits right into the bigger picture of Industry 4.0, which focuses on smart automation, analyzing data, and connecting systems seamlessly. For example, research by Lee and colleagues in 2014 showed that using data-driven diagnostics and forecasts can boost operational efficiency by more than 25%.

This project aims to build a smart predictive maintenance system designed specifically for manufacturing settings. By harnessing data analytics, the system will catch potential machine issues early, helping maintenance teams make smarter, timely decisions. The focus is on making the solution affordable, scalable, and easy to integrate with existing factory setups, especially for small and medium-sized businesses that often face challenges with high startup costs and limited technical know-how.

1.2 Problem Statement

In Malaysia's manufacturing sector, many companies still rely on outdated maintenance strategies either fixing machines only after they break down (reactive maintenance) or servicing them on fixed schedules (preventive maintenance). These approaches often cause unnecessary costs, production delays, and reduced equipment lifespan. This project addresses three main problems:

i. **Unplanned Downtime Due to Reactive Maintenance Practices**

Reactive maintenance is where machines are repaired only after a breakdown frequently leads to unexpected production stoppages, material waste, and higher operational costs. Without real-time diagnostic tools, it is difficult for maintenance teams to detect early warning signs of failure. In Malaysian SMEs, this issue is common due to the lack of affordable predictive maintenance technologies.

ii. **Inefficiencies in Current Preventive Maintenance Schedules**

Reactive maintenance is where machines are repaired only after a breakdown frequently leads to unexpected production stoppages, material waste, and higher operational costs. Without real-time diagnostic tools, it is difficult for maintenance teams to detect early warning signs of failure. In Malaysian SMEs, this issue is common due to the lack of affordable predictive maintenance technologies.

iii. **Lack of Accessible, Scalable Predictive Solutions for SMEs**

Large corporations like PETRONAS and Intel Malaysia have successfully adopted predictive maintenance through AI and IoT technologies. However, most Malaysian SMEs face financial and technical barriers that prevent them from implementing similar systems. Current predictive solutions are often vendor-specific, costly, or require advanced infrastructure, leaving SMEs

without accessible alternatives.

1.3 Objectives of the Project

Based on the identified problems, this project aims to achieve the following objectives each designed to directly address a specific issue from the problem statement:

i. **Design a Smart Predictive Maintenance System Using Supervised Machine Learning**

To solve the issue of unplanned downtime, the project will create a predictive model that analyzes machine condition data such as vibration, temperature, and load. The model will forecast potential failures before they occur, allowing maintenance teams to act proactively.

ii. **Validate System Accuracy Through Simulated Industrial Data**

The system will be put through its paces using real or simulated sensor data from machines to see how well it performs under various fault scenarios. We'll look closely at important measures like how accurate its predictions are, how often it gives false alarms, and how quickly it responds. This testing is essential to make sure the system is reliable and effective at catching problems before they happen, even in complex industrial settings.

iii. **Implement a Cost-Effective and Scalable Predictive Framework**

To address the affordability issue faced by SMEs, the project will design a lightweight and scalable predictive system architecture. It will utilize open-source tools and modular components to ensure ease of deployment and lower implementation costs.

1.4 Significance of the Project

The development of a smart predictive maintenance system provides several important benefits to the manufacturing sector, including:

i. Reduced Downtime and Increased Operational Efficiency

By predicting failures ahead of time, the system helps maintenance teams stay one step ahead, cutting down on surprise breakdowns and keeping machines running longer without interruptions. This means production stays steady, resources are used more efficiently, and overall productivity gets a solid boost.

ii. Cost-Effective Solution for Small and Medium Manufacturers

Unlike big commercial predictive maintenance systems that can be costly and hard to tailor, this project offers an affordable, flexible, and easy-to-scale solution. This way, small and medium-sized businesses can adopt smart maintenance methods without needing a huge budget or technical expertise. Plus, the system helps bring consistency to maintenance routines across different production lines.

iii. Encouragement of Digital Transformation in Traditional Manufacturing

Using this system helps manufacturing companies take a step toward digital transformation, especially those that haven’t fully jumped into Industry 4.0 yet. It brings factory workers into the world of data-driven maintenance, shifting them from relying on manual checks to making smarter decisions based on automated, real-time insights.

1.5 Scope of the Project

This project focuses on developing a predictive maintenance system tailored to small and medium-sized manufacturing companies in Malaysia. The scope of this project includes:

1.5.1 System Data Collection and Preprocessing

The system will collect sensor or simulated data related to machine performance such as vibration, temperature, and pressure. This data will then be cleaned and preprocessed to remove noise and ensure quality input for machine learning.

1.5.2 Machine Learning Model Development

A supervised learning model such as Random Forest or Support Vector Machine (SVM) will be trained to recognize abnormal machine behavior and predict potential faults in advance.

1.5.3 System Testing and Validation

The system will be evaluated using benchmark or simulated datasets (including local industrial data where possible). Evaluation metrics such as accuracy, precision, and recall will be used to measure reliability and overall system performance.

1.6 Chapter Summary

The chapter starts by discussing the design of a smart predictive maintenance system which aims to reduce unplanned downtimes, optimise maintenance procedures, and provide an affordable maintenance system designed for small and medium enterprises (SMEs). The chapter analyses the weaknesses of reactive maintenance and predetermined scheduled maintenance, and advances to the need for an economically scalable and integratable predictive maintenance system designed for small manufacturers.

The problem statement summarises the challenges with the reactive and time-based maintenance methods, the inefficiencies with fixed maintenance schedules, and the absence of an economically viable and integratable adaptable predictive maintenance solution for small and medium-sized enterprises (SMEs). The way to overcome these challenges includes three objectives which are the development of a predictive model using supervised machine learning, the development of a simple dashboard which provides an interface to monitor machine activities in real time, and the evaluation of the system performance in a simulated environment.

This chapter provides reasons why this project is relevant in the real world and in academia. The primary objective is to improve operational efficiency by closing the digital gap, while presenting a solution that is economically viable for low-resource manufacturers. The main focus of the chapter, which is also the project scope, is explained as the development of the predictive maintenance algorithm, user interface design, and system testing, stating the reasonable assumptions and outlining the limitations that may affect the outcome.

Giving only an overview of the motivations, aims, and scope, this chapter has set the primary groundwork. A context has been set to explain why a smart, scalable predictive maintenance system is needed in the field of manufacturing. This chapter is going to explain in detail the foundational theories, contemporary

technology, and relevant research that are essential to the development of the system.

CHAPTER 2

LITERATURE REVIEW

2.1 Introduction to the Topic

The objectives of this project involve the integration of artificial intelligence (AI) with the automation of the industrial sectors, aimed at creating systems for predictive maintenance (PdM) utilising machine learning (ML) and the Internet of Things (IoT). The manufacturing sector is currently in a major transition that is being referred to as the fourth industrial revolution (Industry 4.0). It revolves around the use of intelligent and automated systems, cyber-physical systems, and sophisticated analytics. During this transition, predictive maintenance has become instrumental in improving the reliability of factory equipment, and minimising the unanticipated downtime and maintenance of factory systems.

In the maintenance of complex and advanced production lines, the use of old maintenance methods such as reactive maintenance (only fix the machines when they break) and preventive maintenance (scheduled balancing of routine services) has become unacceptable. These maintenance methods often result in high-cost breakdowns, unplanned maintenance, and lost production (Mobley, 2002). Predictive maintenance, on the other hand, enables organisations to make effective and efficient data-driven decisions in the prediction of equipment failures. This is achieved through the identification of patterns and anomalies in machine data that are collected in real-time.

Current developments in machine learning in conjunction with the Internet of Things have made it possible to implement intelligent Predictive Maintenance (PdM) systems in manufacturing systems. Sensors placed in machines have the ability to collect vast amounts of data on vibration, temperature, sound and load situation. This data is analysed with ML algorithms to detect wear, deterioration, and fault prognostic indicators, enabling maintenance to occur before these issues arise.

(Zhang et al. 2019)

The potential cost savings described above is just a part of the value of this technology. In a 2018 McKinsey & Co. report, it was noted that AI-based Predictive Maintenance has the ability to decrease downtime of machines by 50%, decrease maintenance costs by 10-40%, and decrease the requirement for new equipment by 3-5%. The situation is even more relevant to Small and Medium Enterprises (SMEs) where the budgets are constrained and the equipment breakdown poses a larger risk. Unfortunately, this has made it difficult for smaller manufacturers to implement these systems as the commercial PdM systems are expensive and difficult to modify.

The purpose of this project is to begin development of the predictive maintenance system that fills gaps in the industry. The goal of this customisable system is to use supervised machine learning on simulated industrial data to forecast machine malfunctions in real-time. The aim is to use supervised machine learning on simulated industrial data to forecast machine malfunctions in real-time, but adaptable across numerous machines, scenarios, and industries. To cater to those that do not have advanced technical training/experience, the systems will include a design that will optimise user interaction and engagement to facilitate understanding of predictive maintenance.

The incorporation of Artificial Intelligence and Internet of Things technologies into Predictive Maintenance has the potential to transform the manufacturing industry. Intelligent predictive maintenance systems have also established the baseline threshold for systems in the industry in the current market. The current industrial landscape is predicated on systems that optimise efficiency, equipment uptime, and provide data for strategic industrial planning for operational advantage and sustainability. This project assists the industry in achieving its targets by implementing a predictive maintenance system that is a comprehensive solution based on empirical research and is ready for deployment in the field.

2.2 Literature on Predictive Maintenance Approaches

2.2.1 AI-Driven Predictive Maintenance and Machine Learning Algorithms

The rise of Artificial Intelligence and machine learning integration into modern industrial systems has allowed Predictive Maintenance (PdM) to move from simple time-based strategies to more complex condition-based and predictive approaches. Through the use of time-series data and historical information from machine data controllers (MDCs) and Internet of Things (IoT) devices on machinery, machine learning-based models can accurately predict failures of manufacturing equipment. The use of AI and IoT to investigate more precise, real-time data enhances the capability of failure predictions to identify patterns unlikely to be visible by human operators.

Random Forests (RF) and Support Vector Machines (SVM), both supervised learning techniques, are prominent model classes employed in predictive maintenance in the manufacturing industry. These techniques predict classification based on the input features and machine behavior, such as vibration, temperature, and load. At training time, the RF algorithm, which is one of the ensemble learning techniques, creates several decision trees and predictions are aggregated at the leaves based on the mode of the classes (classification) or average value (regression). This method is highly resistant to overfitting and is thus more suitable given the noisy and incomplete datasets, a feature commonly found in most real-life data from industrial settings. As noted in Tsallis et al. (2025), the ability of Random Forests to handle large datasets, as well as the data variability, qualifies them as one of the most important techniques in PdM systems, particularly within IoT-enabled industrial settings.

SVM is also one of the paramount supervised algorithms and works best when the data is high-dimensional, as is the case when one considers the sensor data from multiple machines. SVM operatEs by determining the most optimal hyperplane that segregates the data into multiple classes, such as normal operation

versus failure. Sethi (2023) focuses on SVM's benefits during fault detection when the dataset is highly imbalanced, i.e., when there is a surplus of normal operation instances as opposed to failure instances. The high effectiveness of SVM in high-dimensional spaces regarding the determination of complex and intricate decision boundaries and complex data, along with its proficiency in identifying and resolving complex patterns, innovative and high-impact decision making, and serves the best for applications in predictive data analytics (or predictive maintenance).

Of late, deep learning, particularly Long Short-Term Memory (LSTM) networks, and the more nuanced applications of such networks have been in high demand for the purpose of using time series data as captured from IoT sensors. In the prediction of equipment failure, the learning of sequential data over time is critical, and so LSTM, as a form of recurrent neural network (RNN), is useful. The works of Singh, Jain, and Kamal (2025) clearly illustrate the marked advancements that LSTM networks have over in traditional machine learning models that predict the remaining useful life (RUL) of a given equipment, particularly in the field of applications requiring predictive analytics and other machine learning frameworks that are intended to act on the basis of data that is organized in sequential patterns such as the temperature (or other pertinent metrics) over time (for example, vibration).

PdM machine learning models assist in anticipating asset failures, optimizing maintenance timing, minimizing downtime, and proactively scheduling maintenance activities, allowing maintenance teams to intervene before asset failures and maintain equipment in a functional state. Maintenance intervention timing unshackles operational cost constraints as unplanned downtime is eliminated, a prominent concern for small and medium enterprises (SMEs) plagued by production loss during downtime.

2.2.2 Integrating IoT with Predictive Maintenance

The implementation of Internet of Things Technology has positively impacted predictive maintenance technology. IoT is the technology that connects different machines through the internet. Within a predictive maintenance technology framework, IoT sensors are embedded into different machines to receive real time information about the machines vibration levels, temperatures, and pressure/load statuses. This information is sent to a control center and analyzed to understand if there are any deviations from normal machine behavior that indicates the machine is about to fail.

As described by Pradeep et al, (2024), another benefit of the IoT PdM system is that it supports real time monitoring that in turn provides the system the capability to make intervention and maintenance decision before the machine reaches a failure point. IoT sensors provide a continuous stream of information that allows predictive maintenance the ability to forecast the future health of the machine, and of machine issues that may cause failure to occur. Poor quality data collection through the sensor system during the implementation of an IoT system will significantly hinder the predictive modeling capabilities of IoT systems. Thus, substantial effort is required in the collection, cleaning, and pre processing of the data from the machine learning in order to drive the systems predictive capabilities.

The immediacy of the data is also important for predicting maintenance needs in real time. Data edge computing, which involves servicing data at source – the IoT sensors – is pivotal in mitigating latency in data relay towards to a central system. Sethi (2023) cites edge computing's efficiency in real-time decision making because of hyper local data processing rather than relying entirely on the cloud. This advantage of local computing is vital in situations which quick decision making is needed in order to perform maintenance to avoid a breakdown of equipment.

While the immediacy of processing the real-time data from maintenance prediction sensors is useful, so is the historical data analysis. Such past data from

analytical machines in a predictive maintenance system can identify patterns, such as seasonal breakdown cycles from wear and tear, and operational conditions which trigger the failure of the machines. This analytical process also achieves the optimal maintenance scheduling and servicing of machines within a safe operating time frame that extends the life of the machine and reduces downtimes from maintenance work.

Thus the merging of IoT with machine learning predictive maintenance is the most seamless solution to loss of efficiency from failure in equipment used in the manufacturing process.

2.2.3 Challenges and Limitations

Even if integrating IoT and machine learning within predictive maintenance is advantageous, challenges remain. As Sethi (2023) points out, the integration of legacy systems is a problem when coupling with IoT-enabled predictive maintenance. Much of the IoT-Equipped predictive maintenance solutions for factories have equipment and systems that have been in operation for many years. These older systems do not have sensors, and in many cases, can not interface with IoT networks. Adding sensors to such legacy equipment and systems is expensive and can be technologically daunting - a challenge for smaller manufacturers, especially.

Worse still, the integration of the requisite data is still a problem, especially for small to medium enterprises (SMEs) that lack the disposable capacity to maintain the sophisticated technological underpinnings that accompany IoT and machine learning-based systems. Predicative maintenance systems in the market such as IBM Maximo Predict and Siemens MindSphere can achieve incredible results, however, smaller enterprises have to forgo these systems due to their exorbitant costs. IoT sensors, systems to combine with software, and the predictive maintenance, and other systems that need to be integrated almost always exceed the means of most SMEs.

2.3 Explanation of Key Terms, Terminologies, and Theories

To build a clear understanding of the smart predictive maintenance system, it's important to first define the key terms and theories behind the project. This section dives into the main ideas around predictive maintenance, machine learning, the Internet of Things (IoT), and the technical basics that all come together to support the design and deployment of the system.

2.3.1 Predictive Maintenance (PdM)

Predictive Maintenance (PdM) is the analysis of data to identify abnormal machine behaviour and predict machine failure before it occurs. This diverges from the typical methods of maintenance practised: reactive maintenance which involves fixing equipment after it breaks, and scheduled preventive maintenance. PdM monitors equipment in real-time and maintains only when evidence suggests a decline in functionality or when a failure is imminent (Mobley, 2002). The objective is to achieve optimal maintenance timing, reduce unexpected machine downtime, and improve machine longevity.

2.3.2 Condition Monitoring

Condition monitoring refers to encompassing a range of monitoring machine performance metrics, including vibration, temperature, noise, and electrical current. This data flow, enabled by IoT sensors, is transferred to the predictive system. Over time, the system learns to detect the patterns that precede a failure, which allows maintenance to control failure proactively (Jardine et al., 2006).

2.3.3 Internet of Things (IoT)

The Internet of Things (IoT) refers to a system of interconnected and interrelated physical objects that collect and transmit data using sensors, software, and communication technologies. In predictive maintenance, IoT enables the

collection of real-time data from machines and its provision to predictive algorithms. It allows for remote monitoring and consolidated diagnostic capabilities, facilitating faster response times and operational transparency (Lee et al. 2014).

2.3.4 Machine Learning (ML)

Being a subdivision of artificial intelligence, machine learning (ML) entails systems learning from data and improving themselves over time without overt programming. In predictive maintenance, ML models are trained from historical labelled data on how to identify patterns that result in failures. Once this training phase is over, the models can, and will, leverage their knowledge to make predictions on the 'issues' contained in new data before such issues emerge. Supervised learning is particularly predominant in PdM, where the models learn from the specific cases of failures documented as input-output pairs (Zhang et al., 2019).

2.3.5 Supervised Learning Algorithms

Supervised learning means training a model using a dataset that includes both input features like temperature and vibration, and the corresponding output labels, such as whether the machine is operating normally or experiencing a fault. Common supervised learning algorithms used in PdM include:

- **Decision Trees**: Used for classification tasks, providing clear if-then logic for predictions.
- **Random Forests**: An ensemble method that improves decision tree performance by reducing overfitting and increasing accuracy (Breiman, 2001).
- **Support Vector Machines (SVM)**: Effective for binary classification, especially when the data is high-dimensional.
- **Artificial Neural Networks (ANNs)**: Modeled after the human brain, ANNs can detect complex nonlinear relationships in data and are useful for signal-

based fault detection.

2.3.6 Statistical Pattern Recognition

A large number of predictive models study past data in order to recognise statistical patterns, then implement what they've learned into future data. Learning models make predictions based on feature extraction, which is the process of transforming noise sensor signals into relevant data i.e. peak value, frequency components, and kurtosis. In this way, particular predictions are made (Lei et al., 2018).

2.3.7 Prognostics and Health Management (PHM)

While keeping in mind what PHM is, it is about most precisely predicting how long an asset will last and estimating when maintenance will be most effective. Combining real-time tracking, diagnostics, and forecasting (PHM) aids in improved maintenance scheduling. Predictive PHM systems often employ AI not only to predict machine failure, but when it will occur (Heng et al., 2009).

The premise of this project is, first of all, the supervised machine learning algorithms, the training of which is based solely on condition monitoring data, and whose operation is made possible by IoT systems for data accumulation and transfer. These components combined offer an excellent foundation for the ML-based predictive maintenance system proposed in this study.

2.4 Existing Solutions and Technologies

Many predictive maintenance systems have been created, both in academic studies and commercial industries, using different technologies and methods to predict equipment failures and boost efficiency. This section offers an overview of these existing systems, compares their strengths and weaknesses, and points out the gaps that the proposed system seeks to fill.

2.4.1 Review of Current Systems

1. IBM Maximo Predict

IBM Maximo Predict is an enterprise-level predictive maintenance solution that uses machine learning and analytics to foresee equipment failures and enhance asset performance. It works seamlessly with IBM's other enterprise management tools, offering insights into asset health through real-time data analysis (IBM, 2020). Maximo Predict gathers data from various sources, including sensors, maintenance logs, and past performance records, to forecast potential failure points and suggest corrective actions.

Strengths:

- Strong integration with IBM’s other software, allowing comprehensive asset management.
- Cloud-based, which makes it scalable and suitable for large enterprises.

Weaknesses:

- High cost, making it less suitable for SMEs.
- Complex setup requiring technical expertise to integrate with existing systems.

2. Siemens MindSphere

Siemens MindSphere is a cloud-based industrial IoT platform that makes it easy to collect, analyze, and visualize sensor data from manufacturing equipment. It supports predictive maintenance by providing tools to monitor machine health and performance in real time. With the help of machine learning algorithms, MindSphere can spot anomalies and predict potential failures before they happen (Siemens, 2021).

Strengths:

- Offers advanced analytics and machine learning capabilities.
- Supports a wide range of industrial sensors and equipment.

Weaknesses:

- Its reliance on cloud computing introduces latency in real-time decision-making.
- The system can be complex to deploy in existing manufacturing environments without substantial IT infrastructure.

3. SAP Predictive Asset Insights

SAP Predictive Asset Insights is another cloud-based platform designed to improve asset performance management. It uses machine learning and data analytics to predict equipment failures and recommend the best times for maintenance. This solution is especially beneficial for large manufacturing companies and integrates seamlessly with other SAP enterprise tools (SAP, 2020).

Strengths:

- Strong integration with the SAP ecosystem.
- Provides predictive insights based on historical and real-time data.

Weaknesses:

- Expensive, particularly for small to mid-sized enterprises.
- Requires specialized knowledge to integrate and use effectively.

2.4.2 Comparison of Technologies

The primary technologies used in existing PdM systems include machine learning models, cloud computing, and IoT sensor networks. Here is a comparison of some popular approaches used in predictive maintenance:

Table 2.1: Comparison of Technologies

Technology	Advantages	Disadvantages
Machine Learning	Accurate fault prediction, scalable, data-driven	Requires large datasets, complex model training
Cloud Computing	Scalable, centralizes data, supports advanced analytics	Data privacy concerns, latency issues in real-time systems
IoT Sensors	Continuous data collection, real-time monitoring	Sensor failure risk, high deployment costs for large setups
Edge Computing	Reduces latency, local data processing	Requires local computing resources, not as scalable

2.4.3 Gaps in Existing Solutions

While existing predictive maintenance (PdM) systems have seen success, there are still several challenges, especially when it comes to smaller manufacturers and less standardized environments. Some of the main limitations include:

- **Cost and Scalability:** Many of the current systems, like IBM Maximo and SAP Predictive Asset Insights, are built for large enterprises and require heavy financial and technical investment. Smaller companies often can't afford these systems, and their complexity can make them hard to adopt (Lee et al., 2014).
- **Integration Challenges:** Commercial systems usually need complex

integration with existing enterprise and IT infrastructure. For small and medium-sized enterprises (SMEs), the cost and time required for integration are too high, making it difficult to take advantage of advanced PdM technologies (Moses et al., 2022).

- **Real-Time Data Processing:** While cloud-based solutions offer scalability, they often come with delays in processing real-time data, which can be a big issue in industrial settings where quick maintenance decisions are needed. Edge computing, which processes data closer to its source, could help with this, but it's not widely used in PdM systems due to the need for local computing power (Zhao et al., 2020).
- **Customization:** Many commercial PdM solutions are designed for specific machines or industries. There's a gap for a more flexible, general solution that can easily be adapted to different manufacturing environments (Lei et al., 2018).

These challenges highlight the need for a cost-effective, scalable, and easy-to-use predictive maintenance system that can be easily implemented across various industrial settings, particularly in small and medium-sized businesses. The proposed system aims to address these issues by offering a modular, adaptable solution focused on reducing costs and simplifying integration.

2.5 Relevant Algorithms and Methodologies

Predictive maintenance systems depend a lot on data-driven algorithms to analyze how machines are running, spot any unusual behavior, and forecast possible failures. In this section, we'll take a closer look at the main machine learning algorithms used in predictive maintenance, how they're put into practice, and the common metrics used to measure how well they work.

2.5.1 Survey of Relevant Algorithms

Several machine learning algorithms have been applied to predictive maintenance tasks, each with its strengths and weaknesses. The most commonly used algorithms include:

1. **Decision Trees**

   The employed method of splitting the dependent variable (DV) for supervised learning models is often done according to the most important features; hence, a decision tree is formed. Due to their representation being tree-like, decision trees are easy to construct, understand, and visualize. This is of great benefit to industry as managers often require the model predictions to be justified. In predictive maintenance systems, decision trees are employed to classify machine state (normal, warning, or failure) according to the sensor data.

   Advantages:

   - Easy to understand and visualize.
   - Fast to train, with clear decision rules.

   Disadvantages:

   - Prone to overfitting, especially with noisy data.
   - Can struggle with complex, high-dimensional data (Breiman, 2001).

2. Random Forests

Random Forests are an extension of the functionality of Decision Trees in the sense that they use an ensemble technique where they split the data into multiple subsamples and train different trees in the ensemble on each subsample and then combine the predictions made by the trees in the ensemble in order to construct a model that is accurate and more reliable than using a single decision tree. In this case the technique will help reduce overfitting and increase the general predictive power of the model. Random Forests are particularly efficient in predictive maintenance because of their ability to process large and complex datasets and their robustness to noise in data.

Advantages:

- High accuracy and robustness.
- Can handle a large number of features and handle missing data well.

Disadvantages:

- Difficult to interpret due to the ensemble approach.
- Computationally expensive with large datasets (Breiman, 2001).

3. Support Vector Machines (SVM)

Supervised learning algorithms that find the best hyperplan that separates the data into different classes are called Support Vector Machines (SVM). SVM are commonly used in predictive maintenance for binary classification problems such as distinguishing between healthy and faulty machines. It works well in the presence of complex data and high dimensions. SVM are effective in cases where data may not be linearly separable as well due to the applications of kernel tricks.

Advantages:

- Effective for high-dimensional data.

- Works well for both linear and non-linear classification tasks.

Disadvantages:

- Sensitive to the choice of kernel and parameters.
- Requires more computational resources for large datasets (Cortes & Vapnik, 1995).

4. Artificial Neural Networks (ANNs)

Deep learning models like the ANN are highly capable of performing complex analyses of sensor data owing to the presence of several layers of interconnected 'neurons' responsible for processing data. In predictive maintenance, ANNs are able to detect non-linear relations because many machines are able to fail, and their components are able to interact in complex addtively. In recent years, models like the Convolutional Neural Network have been able to analyse and perform fault detection using pathe data like vibrations and acoustic signals.

Advantages:

- Capable of modeling complex, non-linear relationships.
- Effective for large-scale and unstructured data.

Disadvantages:

- Requires a large amount of data to train effectively.
- Computationally expensive, especially for deep networks (Zhang et al., 2019).

5. K-Nearest Neighbors (KNN)

The K-Nearest Neighbors algorithm is relatively simple compared to most algorithms. It assigns a class to a data point based on whichever class is the most frequent in its nearest neighboring data points. K-Nearest Neighbors is used in predictive maintenance to see if a machine is operating normally or if it

is showing signs of a fault by comparing it to past similar cases.

Advantages:

- Simple to implement and interpret.
- No explicit training phase; works well with small datasets.

Disadvantages:

- Computationally expensive during the prediction phase.
- Not suitable for high-dimensional data (Cover & Hart, 1967).

2.5.2 Implementation Approaches

There are key steps involved in the implementation of systems of predictive maintenance that include the collection of data, the cleaning of data, the preparation of data, the extraction of useful features, the training of the model, and the deployment of the solutions. Generally, these steps occur in the following order:

1. **Data Acquisition and Preprocessing**

   Installing a predictive maintenance system begins with collecting data from the machine's sensors. Data is typically recorded in the form of time-series signal data. Before we can utilize it though, there is a lot of raw, unfiltered data that needs to be reviewed, pre-processed, and cleaned (removing noise, value normalization, and outlier detection). All of these steps must be taken to ensure that the data that goes into the machine learning models is ready for thorough analysis.

2. **Feature Extraction**

   Before implementing any machine learning technique, every algorithm needs refined input data, which are the features that represent the data effectively, and in the case of raw sensor data, the features represent the sensor data effectively. The most common strategies for features extraction include calculating statistical metrics such as mean and variance, analyzing the

frequency components calculating the Fourier Transform, and discovering time-based patterns with autocorrelation. The extracted features are the most representative so that the algorithms can predict possible failures.

3. Model Training and Validation

Once the data cleaning and feature extraction processes are completed, machine learning models are trained on labeled datasets. In the training process, the model adjusts its parameters until it can make accurate predictions. The validation stage tests the model s predictions on a different dataset than the one used during training. To reduce the risk of overfitting, techniques such as cross-validation are used to estimate the efficacy of the model.

4. Real-Time Monitoring and Prediction

Following the completion of the system development life cycle, the predictive maintenance system will move to the operational phase and continually execute streaming analytics to determine the state of each piece of equipment. The system receives new sensor data from equipment, and the model predicts the state of failure of the equipment: a likely failure state or a functioning state.

2.5.3 Evaluation Metrics

To assess the performance of predictive maintenance models, several evaluation metrics are commonly used:

- **Accuracy:** Measures the proportion of correctly classified instances.
- **Precision and Recall:** Precision tells us how many of the predicted positive cases were actually correct, while recall shows how many of the real positive cases the model was able to catch. Both are important for understanding how well the model can spot faults without raising too many false alarms.
- **F1-Score:** The harmonic mean of precision and recall, used when there is a need to balance both metrics.
- **Root Mean Squared Error (RMSE) and Mean Absolute Error (MAE):** Used in regression models to evaluate the accuracy of remaining useful life (RUL) predictions.
- **Confusion Matrix:** A table used to evaluate classification models, providing a more detailed performance breakdown.

2.6 State of the Art in the Field

Some of the recent advancements made in predictive maintenance (PdM) are the results of improvements made in machine learning, the Internet of Things (IoT), and data analytics. Since there are more sensors and more powerful machine learning tools, it has become easier than ever before to deployed and fine-tuned machine learning. This section highlights the most recent advancements in the field of predictive analytics, the most recent trends, and the expected future developments.

2.6.1 Recent Advances

1. **Federated Learning for Predictive Maintenance**

   Machine learning has recently developed a new concept called federated learning which allows models to be trained on several devices or servers at once and eliminates the need to share confidential information. It benefits predictive maintenance soutions for industries that require high data privacy and security. federated learning helps build privacy-preserving PdM solutions while keeping the costs down that would normally be spent in data transmission, and still being able to learn from the data at rest on multiple devices (McMahan et al., 2017).

   Applications in PdM:

   Used to train predictive models on equipment data collected at different factory locations without transferring the data to a central server.It helps industries comply with data privacy rules by keeping sensitive operational data on-site, while still allowing them to gain valuable insights through machine learning.

2. **Edge Computing in Predictive Maintenance**

   Edge computing is when the data is processed at the source with the aid of the network's edge instead of transferring everything to the cloud. This allows for faster real-time decision-making which is required in predictive maintenance

systems to avoid costly breakdowns stemming from delayed actions on equipment issues. With the adoption of more edge devices, PdM systems can be implemented directly on the devices with reduced reliance on cloud services.

Applications in PdM:

Real-time failure prediction by processing data locally on IoT-enabled machines.Provides fast response times and reduces dependency on centralized systems, especially in environments with limited connectivity.

3. Explainable AI (XAI) in Predictive Maintenance

XAI is a subsection of artificial intelligence concerned with facilitating human comprehension of machine learning models. Models such as neural networks are powerful and yield impressive predictions, however their “black-box” nature makes it difficult for maintenance teams to trust and understand their predictions. Recent developments in XAI such as LIME (Local Interpretable Model-agnostic Explanations) and SHAP (Shapley Additive Explanations) tools illuminate the decision-making processes of complex models, enabling users to build trust and act upon the insights.

Applications in PdM:

Provides maintenance workers with clear explanations of why a particular machine failure is predicted and improves the adoption of PdM systems by increasing transparency, especially in high-stakes environments like manufacturing plants.

2.6.2 Trends and Future Directions

1. Digital Twins in Predictive Maintenance

A Digital Twin represents a virtual version of a real-world physical object, system, or process, and replicates it for the purpose of running real-time analysis and predicting behavior and outcomes into the future. With the purpose of predictive maintenance, Digital Twins are able to track the health of a

machine by integrating real-time sensor data with machine simulation and processing. This method provides a unique understanding of the behavior of the equipment across a variety of conditions and, when combined with machine learning, can predict future failures with a high degree of accuracy.

Future Impact:

Digital twins will allow for highly accurate simulations of machine behavior, helping to predict issues before they manifest in the physical world. They will also enable optimization of maintenance schedules and provide deep insights into asset performance across their lifecycle.

2. **Self-Healing and Autonomous Systems**

Predictive maintenance systems will continue moving toward self-healing and self-autonomous maintenance whereby they will not only sense an issue but will repair it without human intervention. These systems will use an amalgamation of machine learning, IoT, and robotics to detect and evaluate a fault, and perform rudimentary autonomous repair.

Future Impact:

Increases operational efficiency by reducing downtime and human intervention. Reduces the reliance on skilled personnel for maintenance tasks, making maintenance systems more autonomous.

3. **AI-Driven Predictive Maintenance for Non-Traditional Assets**

AI-driven predictive maintenance is no longer confined to the old-world industrial machinery; it is growing to include the likes of electric vehicles, alternate energy technologies such as wind and solar, and even consumer electronics. These newer assets present particular challenges around their complexity, disparate data streams, and different operational parameters. Hence, the focus of the research is increasingly around the construction of models to capture and predict the failure and to tune performance for such systems.

Future Impact:

Expands the application of predictive maintenance to emerging sectors, broadening the scope and impact of AI-powered PdM systems. Creates more opportunities for sustainability through better- maintained renewable energy systems and reduced waste in manufacturing.

2.6.3 Emerging Technologies Shaping the Future of PdM:

- **Quantum Computing**: By facilitating the processing of larger and more intricate datasets at an accelerated pace, quantum computing continues to progress toward the ability to reshape predictive maintenance. In the future, quantum computing will incorporate more innovative algorithms, such as those designed to optimize and detect anomalies, greatly enhancing the capabilities of predictive maintenance (Farhi et al., 2014).
- **Blockchain**: Blockchain technology might improve PdM by providing a decentralized and almost unchangeable record of a piece of equipment's performance and maintenance history. Since PdM systems need complete and precise historical maintenance data, ensuring data honesty and transparency is vital (Zohar et al, 2021).

2.7 Critical Review of Similar Products or Systems

In this section, we take a look at existing predictive maintenance systems and their applications in both academic research and industry. We also explore how the proposed system builds upon, improves, or differs from these current solutions. This comparison helps highlight the strengths and weaknesses of existing systems and shows why the unique features and capabilities of our proposed system are needed.

2.7.1 Review of Similar Projects

1. NASA CMAPSS Dataset Models

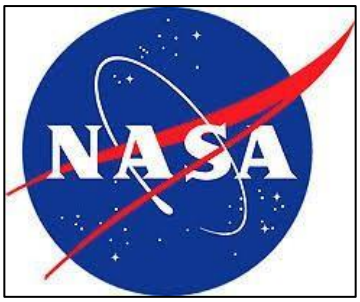


Figure 2.1: NASA Logo

The CMAPSS database contains time series data from aircraft engines, including sensor readings that capture the temperature, pressure, and vibration of the engine over time. This dataset has become one of the standard tools for building and testing models designed to predict the Remaining Useful Life (RUL) of aircraft engines using machine learning, including Support Vector Machines (SVM), Random Forest, and Long Short-Term Memory (LSTM) models. The dataset has been instrumental to numerous studies, including those developed by the NASA Prognostics Center of Excellence (Saxena et al., 2010).

Sub-datasets	FD001	FD002	FD003	FD004
Engine units (training)	100	260	100	249
Engine units (testing)	100	259	100	248
Operational conditions	1	6	1	6
Failure modes	1	1	2	2
Minimum length of testing units	30	21	38	19
Training samples	20631	53759	24720	61008
Training samples with N_{hr}	17731	48819	21820	57522
Testing samples	100	259	100	248

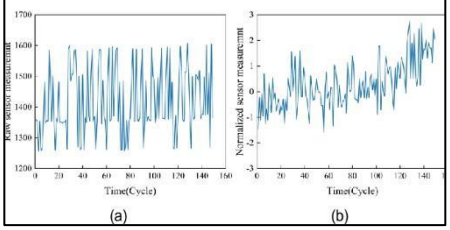


Figure 2.2: Example of NASA CMAPSS Dataset

Strengths:

- The dataset is comprehensive and highly detailed, offering a robust platform for model development and benchmarking.
- It has been extensively used in research, contributing to the standardization of PdM methodologies.

Weaknesses:

- The dataset is specific to aircraft engines and does not generalize well to other industries like manufacturing or automotive.
- The complexity of the data and the focus on RUL prediction make it less applicable to real-time predictive maintenance in non-aerospace settings.

2. SKF Insight

Figure 2.3: SKF Insight Logo

As one of the fabled manufacturers of bearings and seals, SKF has created a cloud-based predictive maintenance solution called SKF Insight. SKF Insight uses IoT sensors to monitor temperature and vibrations of machines and equipment. The system uses machine learning to predict and detect anomalies to foresee when a machine will fail. The system sends alerts whenever an anomaly is detected so incase issues arise, the maintenance teams can resolve the issue before a problem escalates.

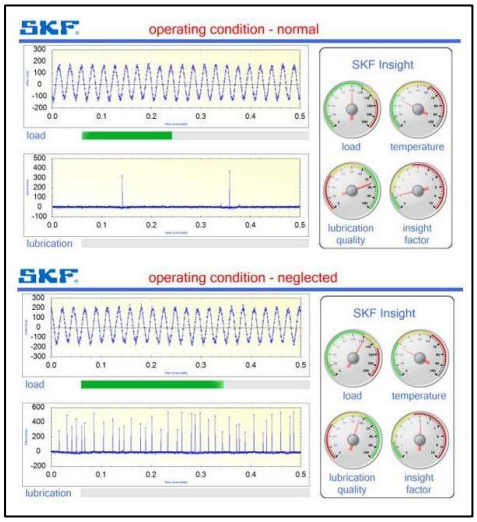


Figure 2.4: Example of SKF Insight Cloud-Based PdM

Strengths:

- Provides real-time monitoring and early fault detection, reducing the need for manual inspections.
- Cloud-based, which allows for scalable deployment across large industrial setups.

Weaknesses:

- The system is tailored for specific SKF products and may not be easily adaptable to other manufacturers' equipment.
- Its reliance on cloud computing may introduce latency and performance issues in environments with limited connectivity.

3. PREDICT-IT Framework

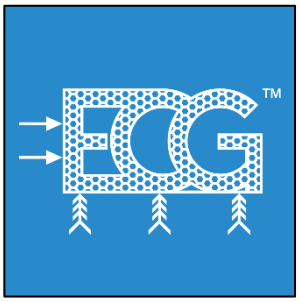


Figure 2.5: PREDICT-IT Logo

Predict-IT employs Random Forests along with other machine learning techniques to forecast motor failures from sensor data within the industrial ecosystem's parameters where continuous monitoring is essential for the seamless functioning of the processes. The specific objective of Predict-IT is to detect motor faults and provide additional actionable information through the user-oriented interface.

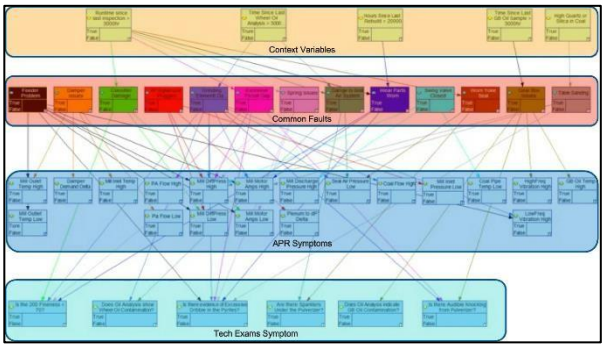


Figure 2.6: Example of PREDICT-IT PdM

Strengths:

- Uses a range of machine learning techniques, allowing flexibility in fault detection.

- Provides a user-friendly interface for plant operators, making it easy to interpret results.

Weaknesses:

- The system focuses mainly on motors and may not be applicable to other types of industrial equipment.
- It requires extensive training data and may not perform well in environments with limited historical data.

2.7.2 How My Project Differs

The proposed predictive maintenance system differentiates itself by focusing on the following unique features:

1. **Generalizability Across Multiple Machine Types**

   Many systems which do exist are tailored to specific machines such as motors, bearings, or engines. Whereas, the adaptive system proposed here is built to suit different types of industrial initial equipment. By utilizing machine learning in the form of Random Forests and SVM, the system can be trained to predict failures in various types of machinery which would make it suitable for a variety of manufacturing environments.

2. **Cost-Effectiveness and Scalability for SMEs**

   Forecasting maintenance solutions such as SAP Predictive Asset Insights and SKF Insight often require specialized infrastructure and are quite expensive. The proposed system aims to be lightweight and scalable, improving accessibility and affordability for small and medium enterprises (SMEs). This will enable smaller manufacturers to access sophisticated predictive maintenance systems without the burden of a large initial investment and with little to no need for IT personnel.

3. **Integration of Real-Time Data with Edge Computing**

   In contrast to system architectures reliant on centralized cloud-based data processing systems, the architecture proposes the use of real-time edge computing data processing at the source of the computing data. This characteristic of the system architecture eliminates potential delays and facilitates the real-time processing of crucial equipment fault detection and equipment malfunction recognition. The system design guarantees edge computing so it can real-time replicate continuous sensor data. This allows immediate feedback, even to maintenance teams, in areas with limited internet connectivity.

4. **User-Friendly Interface for Non-Expert Users**

   One of the primary obstacles to implementing predictive maintenance systems is ensuring that maintenance personnel without technical backgrounds can easily interpret the outcomes and determine suitable responses. To this end, the system includes a user-friendly dashboard that offers straightforward and actionable insights. This approach enhances the system's usability and empowers plant operators with minimal technical skills to embrace the system use with confidence.

2.7.3 Comparison Table

Table 2.2: Comparison Table

System	Machine Learning Models	Real-Time Monitoring	Cloud-Based	Adaptability	Target Users
NASA CMAPSS	SVM,Random Forest,LSTM	Yes	No	Industry-Specific	Aerospace Researchers
SKF Insight	Machine Learning,IoT	Yes	Yes	Limited to SKF gear	Industrial enterprises
PREDICT-IT	Random Forest	Yes	No	Motor-centric	Manufactaring plants
Proposed System	Random Forest,SVM	Yes	Option for Edge	General, Scalable	SMEs,Diverse industries

2.8 Summary of Findings

This section integrates findings from the literature review elucidating thematic overlaps, obstacles, and methods. It also identifies the areas of research deficiency which the proposed predictive maintenance system sets out to address, demonstrating the pertinence and necessity of the undertaking.

2.8.1 Synthesis of Literature

1. **Predictive Maintenance is Critical for Industrial Efficiency**

   Numerous studies have found predictive maintenance to be a crucial tool for improved operational efficiency (Mobley, 2002). With predictive maintenance, unexpected downtime due to equipment failures can be eliminated, as potential failures can be detected before they occur, enabling equipment to endure longer (Mobley, 2002). Additionally, the utilization of predictive maintenance systems in organizations tends to reduce overall maintenance costs, while simultaneously improving the efficiency of operational schedules, reliability of equipment, and overall efficiency of the production processes. This is particularly true for businesses where downtime is directly correlated with decreased income and poor customer satisfaction.

2. **Machine Learning Plays a Central Role in PdM**

   There is a consensus among academics on the role of machine learning models, particularly supervised learning, in predictive maintenance systems (Zhang et al., 2019). Some of the models that are commonly used include Random Forest, SVM, and Neural Networks which are used for anomaly detection, fault classification, and machine health status prediction (Zhang et al., 2019). These models analyze data and learn for the models to be able to predict failures and adapt to future data trends. One of the main reasons that machine learning is so important to predictive maintenance systems is the ability to analyze large quantities of sensor information and recognize complex patterns of failure.

3. **IoT and Real-Time Monitoring are Essential for Data Collection**

   The literature shows that IoT technologies are important for collecting real-time data from machines. Machines have sensors that monitor varying levels of vibration, temperature, and pressure, which allows for predictive maintenance systems to function (Lee et al., 2014). Also, If machines have IoT connectivity, primary data can be stored for central equipment management ensuring remote observation and preemptive actions. However, success of IoT based PdM systems is largely dependent on authentic data and valid sensors.

4. **Current Systems Are Often Expensive and Complex**

   Though predictive capabilities in commercial postpone demand management systems such as IBM Maimo and Siemens MindSphere are advanced, the costs and their integration complexities are significant. The services and systems developed are primarily targeted at large corporations and require big spends on infrastructure and further specialized human resources. Accordingly, there are predictive maintenance systems technologies that are underutilized and that SMEs are unable to adopt (Moses et al., 2022).

5. **Real-Time Processing and Edge Computing Are Emerging Trends**

   New technologies, including edge computing and federated learning, are improving the challenges of latency and scalability of predictive maintenance based on the Cloud Solution. By computing data on the edge within the physical PdM systems are able to make real-time decisions and trigger alerts to maintenance team immediately (Zhao et al, 2020). This fits very well the ongoing need for fast response in environments with high risk of machine failures that can cause serious disruptions of production.

6. **Customization and Generalizability Remain a Challenge**

   As outlined by Lei et al., in many cases, customization is too superficial regarding many contemporary systems. Most of these systems target specific

industries or machine types, making it impossible to adjust them to the needs of manufacturing systems in different environments. Hence, it is of utmost importance to design flexible predictive maintenance systems that can easily adapt to different types of machinery and processes. The system we intend to design does precisely that and will provide the needed versatility to a considerable number of industrial sectors, especially to SMEs that own heterogeneous fleets of machinery.

2.8.2 Research Gap

Based on the findings from the literature, the following research gaps have been identified:

1. **Scalability for SMEs:**

   Most predictive maintenance solutions use too much of the SMEs' (Small and Medium Enterprises) budgets and infrastructure. Many SMEs have limited financial and/or infrastructural capabilities and predictive maintenance (PdM) tools are generally too expensive and complicated for such companies. It is evident that there is a deficit of simple and affordable predictive maintenance systems for small manufacturing companies.

2. **Real-Time Decision-Making:**

   While there are excellent Web-based Analytical tools for Predictive Maintenance, the Analytics some cloud-based systems are spending far too long on analytics, and therefore not providing the analytics in real time. While edge computing is becoming far more popular in the analytical arena, still not widely accepted and utilized in many Predictive Maintenance solutions. This Delay is providing the potential for systems to be developed to Process Analytics on the Equipment, and providing real time insight, and not be reliant on the cloud and the internet.

3. **Generalization Across Multiple Equipment Types:**

   So far, most predictive maintenance solutions have been specialized for particular classes of machines or industries. A more adaptable system that could be tailored to a variety of machines is what is really needed, especially in settings, like small and medium-sized enterprises that have diverse machinces ranging from different producers.

4. **Explainability and Trust in AI Models:**

   Predictive maintenance has been impacted by the ‘black box’ nature of some machine learning models. Non-specialists find it difficult to place trusting confidence within those models. There will always be a growing need to develop the more transparent and explainable AI (XAI) models that provides a clear path for how and why predictions of failure are made. That transparency will allow operators and maintenance people to improve their decision-making using better models.

The gaps showcase the necessity of a cost-effective, scalable, flexible, and easy-to-understand predictive maintenance systems offering real-time insight.

2.9 Chapter Summary

This chapter has reviewed the available literature for predictive maintenance (PdM) concentrating on the main technologies, algorithms, and systems available. This helped in the streamlining of maintenance, reduction of downtime, and increase in operational efficiency. The review detailed how machine learning algorithms, particularly the supervised models like Random Forests, Support Vector Machines (SVM) and Neural Networks, play a critical role in the predicting of failures and the advancing of maintenance techniques to proactive.

We assessed the use of IoT (Internet of Things) sensors to stream machine data in real-time and the emerging use of edge computing for accelerating data processing and decision-making on-site. We briefly reviewed available PdM solutions as IBM Maximo Predict, Siemens MindSphere, and SAP Predictive Asset Insights. These solutions are great, but tend to be expensive and complex, which can deter adoption by small and medium enterprises (SMEs).

Additionally, this review has provided us the gaps in the PdM solutions, such as the unavailability for customization for a variety of machine types, the unavailability for explainable Artificial Intelligence in the predictive models, the lack of scalability to small and medium enterprises and the inadequacies of the current predictive models. There is a solid case for determining a value proposition for a solution that Dr. B is proposing and that is affordable, scalable, easy to use and customizable to a variety of industries.

What this chapter has achieved is to provide the foundations for the proposed system to tackle these challenges to combine the use of machine learning models that are low cost and edge computing to provide a value proposition of real time processing of data with high usability for the non technical user. The next chapter is primarily devoted to building and testing the system, to describe the methodology, to ensure that the value proposition outlined in this literature review has been met.

CHAPTER 3

METHODOLOGY

3.1 Introduction

This chapter explains the methodology used to develop the predictive maintenance system for manufacturing environments. A system designed to reduce unexpected downtime, improve maintenance efficiency, and offer a scalable solution tailored for small and medium-sized enterprises (SMEs), as discussed in Chapters 1 and 2. The main goal is to leverage supervised machine learning algorithms alongside IoT sensors for real-time monitoring and early fault detection, helping prevent equipment failures before they happen and ensuring smoother, more efficient production.

To meet these goals, the V-Model methodology will be the only one used during the system's development. Due to its design, the V-Model methodology will also be the best choice for predictive maintenance. The V-Model methodology focuses on the strong interrelation between each development stage and the validation of that stage. This is vital for systems that incorporate machine learning to make predictions, along with the processing of telemetry in real-time and maintenance planning; all of which require that every element be accurate and reliable, one hundred percent of the time. The V-Model, unlike other exploratory or iterative methodologies, ensures a disciplined and consistent flow of development, which includes risk evaluation and a measurement of the system’s requirements in all successive stages.

In this initiative, the first focus of the V-Model is to determine whether the system requirements are well understood prior to any design or implementation. This is crucial, as predictive maintenance consists of several interdependent subsystems, such as IoT data acquisition, data preprocessing pipelines, anomaly detection models, and user dashboards, that need to work correctly and seamlessly

one unit. The V-Model achieves this by directly associating each development activity with a verification step, which allows the system to reveal any missing pieces, inconsistencies, or erroneous assumptions. The conceptual design of the system then iteratively advances to undergo each of the specialized structured tests, which are aligned with the system's design to determine if there is any divergence in the system's intended purpose and its realized features, to the development of a machine learning model, a database, and the system's interfaces.

Employing the V-Model increases the overall value and reliability of the system as a whole. Predictive maintenance utilization involves estimating telemetry data and forecasting production models. Any predictive errors may result in unproductive system downtime, failure to detect system issues, and/or expensive downtime due to system outages. The V-Model employs top-down planning and bottom-up risk validation to eliminate these threats. Each predictive system component, whether data, algorithms, or visualization, is risk-adjusted to meet program scope baseline objectives. The results of the V-Model foster construction workflow improvements through visibility, resilience, and tracking, which is essential in building a system to interface IoT, machine learning, and cloud dashboards. The approach enables the construction of an effective and reliable Smart Predictive Maintenance System that achieves the program's goals.

3.2 Software Development Methodology

The project will continue in a software development manner in accordance with the V-Model, a planned and stringent method characterized by its correlation between each development phase and its corresponding testing activity. The V-Model is the best method to provide a structured approach to the development of a predictive maintenance to be built, and to ensure that each requirement of the system is validated, verified, and tested. For these reasons this methodology increases the potential of the system to be reliable, performed issue detection, and maintained constant alignment between the requirements, design, and implementation of the system.

3.2.1 Chosen Methodology and Justification – V-Model

Because of the V-Model's highly structured and validation driven approach, which is extremely helpful for development of systems of systems needing precision, meticulousness, and systematic checking like predictive maintenance systems, is used for this Project.

The V-Model focusses on a methodical approach whereby a development stage has an accompanying testing stage. This is done so that any faults in the process are diagnosed earlier and that the requirements of the system are checked continuously. The stages are requirement analysis, system design, model implementation, testing and validation.

V-Model: Structured Verification and Validation

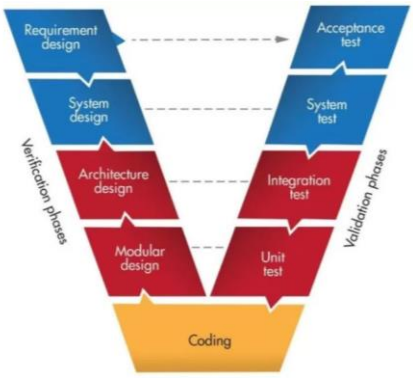


Figure 3.1: V-Model Software Development

With heavy emphasis on verification and validation at all levels of the process, the V-Model thus gets its name from the unique way its phases are organized. Namely, the development phases are on the left, while the corresponding testing and validation phases are on the right, thus forming the shape of a V.

- **Description:** This specific project benefits greatly from having clearly defined system requirements and a dependable, faultless V-Model system. Also, the V-Model is great for predictive maintenance since it ensures that every stage from collecting system data, training the machine learning model and system integrating, gets verified and validated before a new stage in the V-Process is reached.Why it was chosen:

    1. **Rigorous Testing:** Ensuring that machine learning models and IoT sensor data can accurately and consistently identify equipment failures is an iterative process that requires testing and verification. The V-Model offers validation and verification from every tier of development, and allows for orderly confirmation of each component.

    2. **Systematic Development:** Systems for predictive maintenance require many system interconnections, like sensor data collection, data processing, model prediction, and user interface. The stepwise approach in V-Model ensures\that each system is constructed and

tested in an orderly, organized fashion, and verification and validation are done in each step before moving to the next.

3. **Minimized Risks:** As predictive maintenance is key to lose avoidance, the inability to predict maintenance requirements can have dire costs. The V-Model assists the control of the integration of systems to avoid integration issues and make sure all components are working seamlessly together, prior to going live.

- **Validation and Verification in the V-Model**
  - **Verification** is carried out in the processing of data as well as the phases of model training to ensure accuracy in the sequence of steps being undertaken.
  - **Validation** occurs at every stage of the model to guarantee the system's behaviors are consistent with the real-world demands of the manufacturing setting, including precise prediction of faults in real time and subsequent fitting responses.
- **Benefits for Predictive Maintenance:**
  1. Testing at every stage ensures that predictive models are accurate, data flows properly from sensors to models, and the results are practical and actionable.
  2. For example, data preprocessing and feature engineering need to be carefully verified to make sure the input data quality is good enough for training machine learning models.
  3. The system’s overall functionality is validated through simulations and tests to confirm it provides accurate maintenance recommendations and alerts based on real sensor data.

The V-Model has a very systematized sequence of verification and validation

making it very useful in areas of safety-critical and predictive maintenance. This method is particularly useful in increasing the complexity of the system to be reliable. This is especially pertinent in industrial settings where system failures can result in significant impact in money and resources. (Moses et al. 2022).

3.2.2 Step by Step Explanation of Activities Done in Each Phase

The development of the predictive maintenance system uses a hybrid approach, blending the structured steps of the V-Model with the iterative refinement and risk management of the Spiral Model. Below is a detailed breakdown of each phase of the methodology, showing how each part of the project will be carried out.

1. **Planning Phase**
    - **Requirements Gathering:** The project team will collaborate closely with stakeholders including maintenance managers, factory operators, and technical experts to gather detailed requirements for the predictive maintenance system. The main goal is to understand operational challenges like unexpected downtime and define functional needs such as real-time fault prediction, data integration, and a user-friendly interface. This phase ensures the system is tailored to the specific needs of SMEs, addressing both their operational goals and technical constraints (Moses et al., 2022).
    - **Feasibility Study:** This step evaluates whether machine learning models can effectively support predictive maintenance and how they can integrate with IoT sensors. It also includes assessing the technical infrastructure needed, like cloud services for model deployment, and analyzing financial impacts to ensure the system stays affordable and scalable for SMEs (Zhao et al., 2020).

2. **Design Phase**
    - **System Architecture Design:** In this phase, the overall structure of the predictive maintenance system is planned out. This includes key components

like IoT sensors, machine learning models, and a user-friendly dashboard. The architecture is designed with scalability in mind, ensuring it can handle large amounts of sensor data and integrate smoothly with existing factory setups without major disruptions. The design is then verified to make sure all parts communicate effectively and work together to provide real-time predictive insights (Moses et al., 2022).

- **Specification of Verification and Validation Processes:** Since the V-Model stresses validation at every stage, this phase also defines how verification and validation will be conducted throughout development. For example, criteria for validating machine learning models like accuracy, precision, and recall will be set, and data collection methods will be checked to guarantee quality and completeness (Zhao et al., 2020).

3. **Development Phase**

- **Implementation and Integration:** Development starts on both the machine learning models and the user interface. The machine learning models will be trained using historical and sensor data, then integrated with the data collection system. The user interface will be designed to give factory operators clear, actionable insights like fault predictions, machine status updates, and maintenance schedules. Integration between the machine learning models and backend systems will be tested to ensure the predictive maintenance outputs are practical and reliable (Moses et al., 2022).
- **Verification:** Throughout development, each component whether it's the model, data processing, or user interface will be tested against project requirements. This means verifying individual modules work correctly before moving forward. For example, the integration of IoT sensor data will be tested for accuracy, making sure sensor readings are properly fed into the machine learning models to predict faults (Zhao et al., 2020).

4. **Testing Phase**

- **Unit Testing:** Each part of the system like the machine learning models, data collection tools, and user interface will be tested individually to ensure they work correctly on their own. This phase will also check that the system can handle real-time sensor data and generate accurate predictions (Zhao et al., 2020).
- **Integration Testing:** After confirming each component works, the whole system will be tested together to make sure all parts function smoothly when integrated. For example, the data pipeline from collecting sensor data to producing predictions will be tested for seamless data flow (Moses et al., 2022).
- **System Testing:** Finally, the entire predictive maintenance system will be tested in real-world conditions to verify it performs as expected. This includes checking its ability to predict failures using sensor data and evaluating the user interface for usability and accessibility (Zhao et al., 2020).

5. **Evaluation and Deployment Phase**

- **Final Validation:** The system will undergo a final validation to ensure that it meets all functional, non-functional, and usability requirements. This will involve validating the predictive maintenance system under real-world manufacturing conditions to ensure it can predict equipment failures with a high degree of accuracy (Moses et al., 2022).
- **Deployment:** The system will be rolled out in phases across multiple manufacturing lines. The V-Model helps ensure that any issues found during this initial deployment can be quickly addressed, making the transition to full integration as smooth as possible (Zhao et al., 2020).

3.3 Research Methodology

In relation to the predictive maintenance system, this project will employ a quantitative research methodology to collect and analyze data. Questionnaires enable the systematic recording of users' experience, maintenance preferences and challenges. The approach will collect statistically analyzable data and delineate patterns, correlations, and user needs. With regards to the focus of the project which includes the development of machine learning models and the integration of IoT sensors, a quantitative method presents the clarity and reliability necessary to drive the development and design of the system.

3.3.1 Chosen Research Methodology and Justification Why Quantitative Research?

1. **Objectivity and Measurability**

   Quantitative research focuses on the collection of numerical information with the aid of measurement tools such as surveys, which can in turn be evaluated using statistical methods. This guarantees the data are objective and truly measurable. Examples of such measurement tools are Likert Scale (1 to 5) which allows capture of data regarding users' views in relation to satisfaction levels with existing maintenance practices or the value attributed to real-time notifications. Such feedback allows the identification of patterns which lead to the formation of data driven, objective conclusions regarding the potential benefits of the system. Predictive maintenance systems are data driven and therefore the ability to measurement users are needs, users' preferences and the system's performance in the maintenance practices would greatly assist in determining the feature design and priorities.

2. **Statistical Analysis for Large-Scale Data**

   One of the key advantages of conducting quantitative analysis is the capacity to manage the large volumes of datasets for this research project. This

research project seeks feedback from multiple stakeholders, including maintenance managers, operators, and field technicians. To obtain measurable and statically analyzable data, the research has employed the use of primary data collection techniques in the use of questionnaires. Before statistically analyzing the data, the research will collect data from approximately 20-25 participants. This will allow the study to draw conclusions about the data point frequencies in main maintenance problems in the study for example unplanned downtime, and assess the overall satisfaction level the participants had on the maintenance systems in use and the gaps in data prediction systems and data integration systems. More so, quantitative analysis supports the exploration of correlation of variables i.e. the relationship of high rates of downtime to the absence of predictive maintenance practices. Statistics provide more reliable data as opposed to just pure anecdotal observations. This could be more reflective of the realities in the diverse macro settings of the manufacturing sector.

3. **Generalizability and Scalability**

   Quantitative findings have also justified generalizability and the inset outcomes gained from obtaining participant`s structured responses from separate and diverse respondents from a variety of manufacturing sectors. The outcomes from predictive maintenance research developed respondents' structured responses from a variety of sectors. Soft and mid-tier manufacturing sectors have developed diverse predictive maintenance frameworks. The developed frameworks will address other sectors. The predictive maintenance research developed diversified frameworks.

4. **Clear Measurement of Effectiveness and Usability**

   Predictive maintenance systems hinge on a range of measurable metrics which help ascertain the value of the system in predicting failures, ease of

use, and the value of the system in improving efficiency. Predictive systems quantify the metrics to assess how often unplanned downtime occurs, user-staff dissatisfaction with the maintenance methods in use, and the value of system features including predictive alerts, historical data access, and real-time data. Each of these metrics converts user satisfaction, the perceived challenge, and feature priorities to data which assist in sharpening the priorities in a system which aids the project in determining the features of the system which most greatly improve user acceptance and operational efficiency, the impact of which is prediction of user satisfaction and determination of the features data relies on.

5. **Precise User Preferences and Expectations**

   The questionnaire allows us to compute how users value different characteristics of the system, such as the value of getting alerts in real-time, the importance of seeing predictive data, the need for effective data visualization, and the choice of the system's interface as mobile vs. desktop. Since the values assigned to the responses are numerical, it becomes much easier to synthesize the data and determine the most valued features by users, helping the development team focus their efforts, in evidence-based manner, on the features of the interface design. For example, if the most responses are assigned to real-time alerts as an extremely important requirement, that readily signals the requirement is critical enough to warrant its inclusion and focus in further system development to ensure the end product serves the users and the operations.

6. **Clear Identification of Pain Points and System Requirements**

   This study seeks to understand what issues users have with current maintenance practices and how a new predictive maintenance system will be able to address them. For the study to have a solid foundation We need

quantitative data to measure the issues objectively so that things such as the rates of unplanned downtimes and the inefficiencies associated with heavy, un-downtimed, and manual log dominated operations, can be evaluated. When these pain points are quantified, the project can derive system requirements that are both specific and clear This pain-point quantifying system ensures that the most damaging inefficiencies are addressed, such as weak and uneven data systems, low predictive accuracy of failures, and the absence of automated maintenance scheduling so that the maintenance system delivered will be able to meaningfully address and solve multiple maintenance challenges.

7. **Scalability and Flexibility of the System**

   Analyses on the quantitative data provided would further attest the system's functionality on different types of manufacturing environments. For example, the survey would show whether the system should provide support for several machines on different production lines and whether real-time processing is needed by both large factories and small and medium-sized enterprises (SMEs).

3.3.2 Questionnaire Design and Samples

The questionnaire is designed to collect quantitative data from maintenance personnel, factory operators, and technical experts in manufacturing environments. The purpose of this questionnaire is to gather insights on current maintenance practices, challenges faced, and expectations from a predictive maintenance system. The results will help guide the design and development of a system that addresses real operational needs and optimizes maintenance activities.

Form Link:

https://docs.google.com/forms/d/e/1FAIpQLScwsirtN03ilzzil5zQlVx2edlHafRDUEeGOusaJn0O_Vj3PQ/viewform?usp=dialog

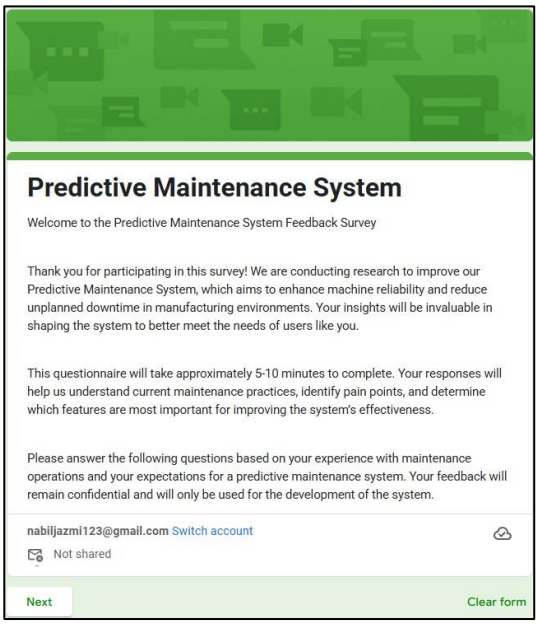


Figure 3.2: Introduction of Survey

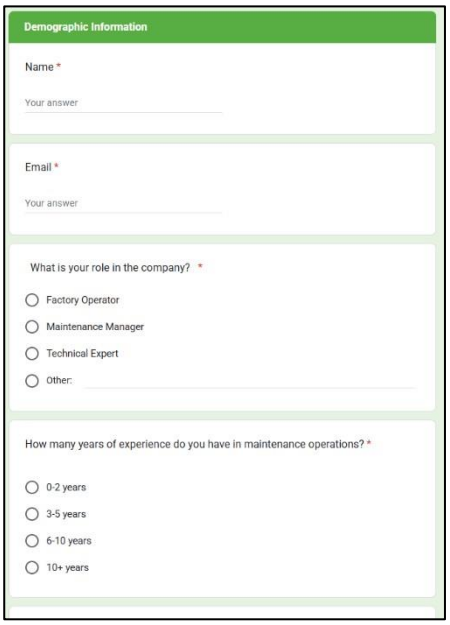


Figure 3.3: Section 1 of Survey

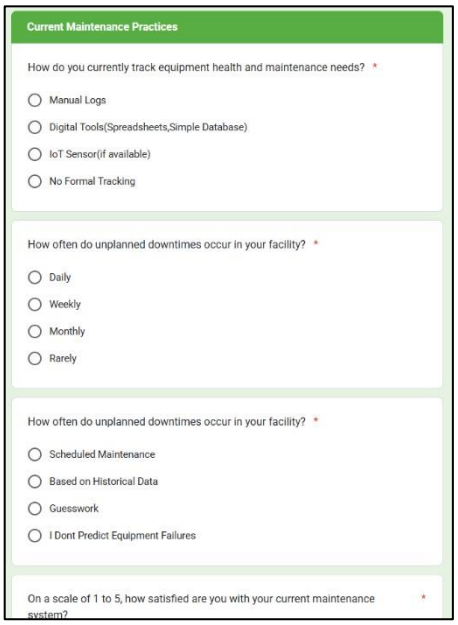


Figure 3.4: Section 2 of Survey

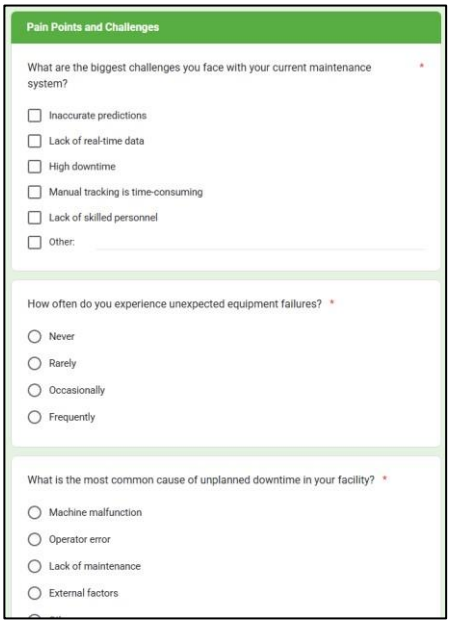


Figure 3.5: Section 3 of Survey

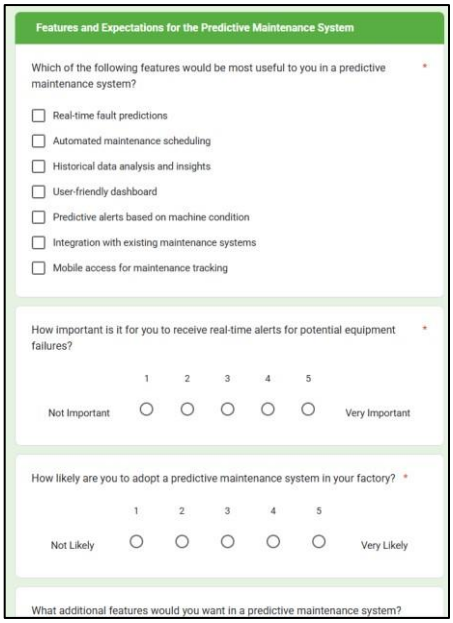


Figure 3.6: Section 4 of Survey

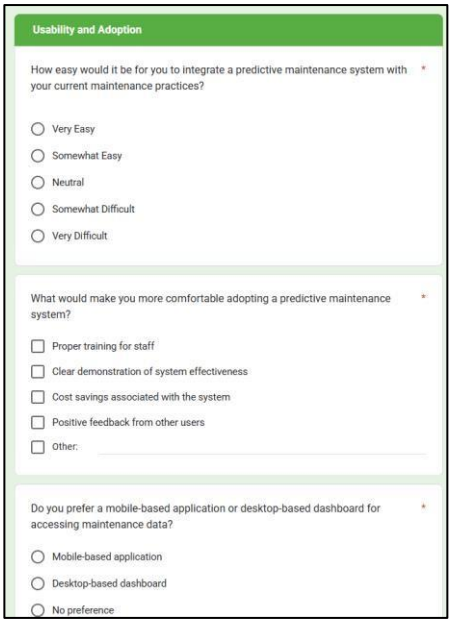


Figure 3.7: Section 5 of Survey

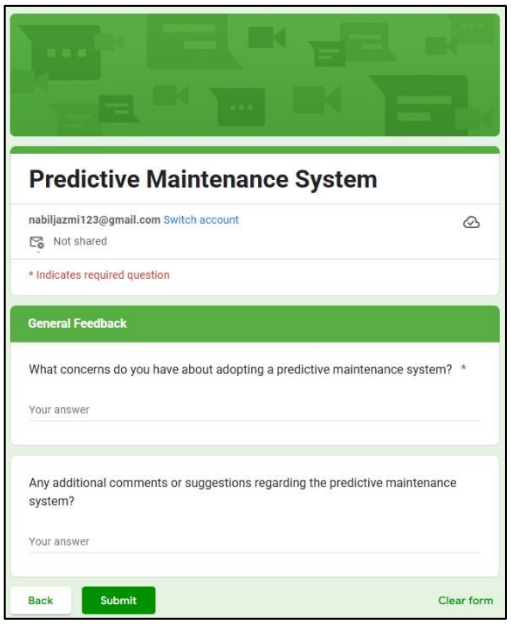


Figure 3.8: Section 6 of Survey

The questionnaire was split into six sections, concentrating on the acquisition of specific user information. This closes with user feedback, hoping to refine the predictive maintenance solution to meet user expectations. It aims to validate existing topics to avoid bias to include prediction maintenance systems using the hypothesis of predictive maintenance. User feedback on the expected simple to use system ensures avoidance of the self-fulfilling bias. It provides final comments, allowing the user to mention anything the system may have overlooked to improve system user congruence. The corrective bias to ensure the simple system is only of

the self-setting bias to provide the real gaps.

The questionaire was directed to 25 users, including maintenance and operators of relevant sectors, automotive, electronics, food, and medicines. Their tiered industrial experience was to ensure the system to be designed for worth of setup and operational diversity. This appriciation of feedback will be of garnished value.

The questionnaire will be disseminated through online channels like Google Forms for participant convenience and efficient collection of responses. The quantitative responses will be analyzed through descriptive statistics in order to isolate and examine the major patterns and preferences. The open responses will be analyzed for their thesauri for patterns and articulating the most prominent concerns to focus on for further system design enhancements.

The findings will inform the system’s requirements and the necessary features to be prioritized and arranged on the user interface to be dedicated to the predictive maintenance system for user utility and efficiency in SMEs

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3.3.3 Analysis of Questionare Data

The responses gathered from 25 participants, including maintenance staff, factory workers, and specialists, provide valuable information regarding participants' maintenance practices in manufacturing environments and customer needs regarding the new predictive maintenance system. The responses provide the primary data to understand the emerging patterns affecting the new system's needs and requirements.

Nearly all the respondents in the data indicate that they are still using either conventional or semi-digitized maintenance. While important participants still used paper and manual mechanisms, some respondents used spreadsheet software or rudimentary software applications. The absence of an automated system raises the question of why there continues to be significant unplanned system downtime. The provided responses indicate that maintenance teams, consistent with literature, also suffer from similar problems of inadequate data recording, poor ability to identify early indicators of machine wear, and reactive maintenance. This illustrates the importance of sophisticated machine learning and predictive analytics to decrease uncertainty and aid in reducing the downtime problems.

From the questionnaire, there is a clear strong interest in the predictive maintenance features. Many respondents noted the importance of receiving alerts in real time, visualizing the data in a meaningful way, and predicting failures accurately and reliably. Many noted that accurate, timely notifications of an impending machine failure would help prevent unexpected machine breakdowns and help reduce production delays. The feedback demonstrates a desire for clear, simple, uncluttered, and easy-to-use dashboards which indicates that usability is a very important factor for such dashboards, especially for operators who may not be very technically inclined. The feedback shows differing preferences in terms of accessibility of the dashboards, with some respondents indicating a preference for desktops and those who appreciate being able to use mobile devices to receive

updates indicating that a flexible, multi-platform type of system would be best.

When respondents were asked to identify their overall pain points, the most frequently mentioned ones were high amounts of downtime, challenges in analyzing maintenance histories manually, and toolsets not being integrated to provide support for proactive maintenance planning. This speaks to the need of having a single cohesive system that captures machine data, performs analysis of the data, and provides actionable insights. The respondents built in system features like automated scheduling, historical trend analysis, and predictive alerting further supports the need of such a system and aligns with the intended capabilities of the system. This is a good confirmation of the direction of this project.

Overall, the questionnaire results show that production staff experience considerable difficulty with the current maintenance processes, and they strongly advocate for the implementation of an artificially intelligent predictive maintenance programme. This feedback also helps direct the prioritization of features for the system, including the needs for greater operational efficiency, wider acceptance, and SME use that determines the need for features such as active tracking, prediction, alerting, and sophisticated data presentation.

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3.3.4 Proposed System Requirement

Table 3.3: Proposed System Requirement

No.	Requirement	Type	Stakeholder
1	The system must collect real-time data from IoT sensors embedded in machinery to monitor equipment health and performance.	Functional	User/Admin
2	The system must use machine learning models to analyze sensor data and predict potential equipment failures before they occur.	Functional	User/Admin
3	The system must send real-time alerts and notifications to maintenance personnel if a potential failure or anomaly is predicted.	Functional	User
4	The system should automatically generate maintenance schedules based on predicted failures, optimizing downtime and maintenance operations.	Functional	Admin
5	The system must store and allow users to analyze historical performance data of machines to help identify trends and optimize future maintenance schedules.	Functional	Admin
6	The system must be scalable, able to handle increasing amounts of sensor data from multiple machines and production lines without significant performance degradation.	Non-Functional	Admin
7	The system must ensure 24/7 availability to accommodate round-the-clock production environments and ensure that predictive maintenance is continuous.	Non-Functional	Admin
8	The system must comply with data security standards to protect sensitive operational data and personal information of users.	Non-Functional	Admin
9	The system must process data and provide fault predictions with low latency, ideally within a few seconds to allow quick decision-making.	Non-Functional	Admin
10	The system must provide a user-friendly interface that allows easy interaction for non-technical users such as maintenance personnel and factory operators.	Usability	User

11	The system should support multiple languages to cater to a diverse user base across different regions and industries.	Usability	User
12	The system should allow customization of features such as alert thresholds, report generation formats, and the frequency of maintenance reminders.	Usability	User/Admin
13	The system should be accessible via mobile devices, allowing maintenance staff to monitor equipment health and receive alerts on-the-go.	Usability	User/Admin

3.4 Proposed System Design

This section presents the proposed design for the predictive maintenance system, which combines machine learning models with IoT sensors to monitor equipment health in real time and predict failures before they happen. The system will give maintenance teams clear, actionable insights to help reduce unexpected downtime and optimize maintenance schedules, leading to better overall operational efficiency.

3.4.1 System Architecture Design

1. This is a user interface visible from a computer. Factory operators and maintenance employees can monitor machine health, get predictive maintenance notifications, and analyze past performance data in real-time. The interface is user friendly offering easy to understand visual signifiers on the health of machines, maintenance due dates, and predictive maintenance recommendations.

2. This layer handles the core system functions. It processes real-time sensor data gathered from IoT sensors attached to the machinery and runs machine learning models to predict possible equipment failures. It also manages communication between the client and database layers, sending real-time updates, alerts, and data to users. Additionally, it oversees system logic to ensure sensor data is accurately analyzed for generating failure predictions and scheduling maintenance.

3. Serving as the backbone of data storage, this layer holds all information collected from IoT sensors and the output from machine learning models. This includes sensor readings, historical performance records, maintenance logs, and future maintenance predictions. The database needs to be scalable to handle growing data volumes from multiple machines and production lines. It must also guarantee security and data integrity, protecting sensitive operational data while allowing authorized users easy access.

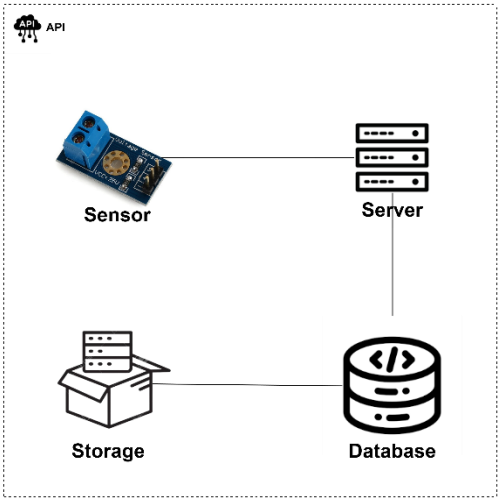


Figure 3.9: System Architecture Diagram

3.4.2 Component Design

1. IoT sensors are the system's starting point for data collection. Installed on machinery, they monitor key performance metrics like temperature, vibration, pressure, and operating speed. They continuously gather and send data to the server for processing. For the predictive maintenance system to work well, these sensors must be reliable, withstand tough industrial conditions, and provide accurate, real-time data to help detect potential failures.

2. Machine learning model, using the collected data, the machine learning model predicts possible equipment failures. It's built from historical data, trained to spot patterns linked to failures, and constantly updated with new information. This model is crucial for making accurate predictions about when and where issues might occur. Depending on the system's needs and sensor data type, algorithms

like decision trees, random forests, and support vector machines (SVMs) are used.

3. Data processing is also a critical part of the system design. Before feeding data into the machine learning models, it must be cleaned, normalized, and pre-processed to ensure consistency and quality. This stage removes noise and formats the data properly for analysis. The data pipeline is designed to handle large volumes of real-time data, making sure predictions are fast and accurate.

4. The UI is designed with ease of use in mind for maintenance teams and factory operators. It's intuitive, allowing users to see real-time equipment status, receive predictive alerts, and generate maintenance schedules based on model predictions. Features include interactive dashboards, data visualizations, and alert notifications, helping operators quickly assess machine health and act to prevent downtime. The UI will also be customizable, letting users adjust alert thresholds, report formats, and other settings to fit their specific needs.

5. The maintenance scheduling tool, which will automatically creates maintenance schedules using the machine learning model's predictions. It helps prioritize tasks so maintenance happens at the best possible times. The scheduling tool will integrate with existing maintenance management systems to keep workflows smooth for maintenance teams.

3.4.3 Data Flow and Communication

The data flow in the predictive maintenance system involves the following sequence:

1. IoT Sensors continuously collect real-time data from machines (vibration, temperature).
2. The data is transmitted to the server layer, where it is pre-processed for noise reduction, normalization, and validation.
3. The machine learning models in the server layer analyze the data and

generate predictions (equipment failure probability).

4. Based on the predictions, real-time alerts are generated and sent to the user interface for factory operators and maintenance managers.
5. The maintenance scheduling component automatically schedules maintenance tasks based on the predicted failure events and integrates with existing systems.



Figure 3.10: Flowchart of PdM System

3.4.4 User Interface Design

The user interface (UI) will be designed to meet the needs of maintenance personnel and factory operators, ensuring that it is easy to navigate, intuitive, and provides the necessary insights. The UI will feature:

1. **Dashboard:**

   A real-time monitoring dashboard displaying the health status of machinery, upcoming maintenance tasks, and real-time alerts for potential equipment failures.Interactive graphs and visual indicators (green, yellow, red) to help operators quickly assess equipment health.

2. **Alerts and Notifications:**

   Real-time alerts to notify operators about equipment issues, upcoming maintenance needs, and other system anomalies.The ability to customize alert thresholds based on equipment criticality.

3. **Maintenance Schedule:**

   A calendar or task list view showing upcoming maintenance activities, including predictive maintenance tasks, and recommendations for when to perform repairs or replacements.

4. **Historical Data Analysis:**

   Data visualization tools (charts, graphs) for reviewing equipment performance trends, identifying recurring issues, and making data-driven maintenance decisions.

3.4.5 Machine Learning Model Design

The predictive maintenance system's machine learning model will be designed to process sensor data, identify patterns, and make predictions about potential equipment failures. The key steps in the design of the model include:

1. **Data Collection:** Gather historical sensor data, such as temperature, pressure, vibration, and operating hours, from previous machine failure instances.
2. **Feature Engineering:** Identify relevant features from the raw data that will help in predicting failures (temperature spikes, vibration anomalies).
3. **Model Training:** Use supervised learning algorithms (such as Random Forests, Decision Trees, or SVMs) to train the model on historical data.
4. **Model Testing:** Validate the model using a separate test dataset to ensure its ability to predict equipment failures accurately.
5. **Model Deployment:** Deploy the trained model into the server layer, where it will analyze real-time sensor data and make predictions about machine health and potential failures.

3.4.6 Testing and Validation Process

To ensure that the predictive maintenance system is both effective and reliable, the system will undergo thorough testing and validation in both simulated and real-world environments:

1. **Unit Testing:** Each component of the system (data collection, machine learning model, user interface) will be tested independently for functionality and accuracy.
2. **Integration Testing:** The components will be integrated and tested as a whole to ensure smooth communication between the IoT sensors, server, machine learning model, and user interface.

3. **System Testing:** The entire predictive maintenance system will be tested using historical data and real-time data from factory machines to assess its performance in detecting equipment failures.
4. **User Acceptance Testing (UAT):** Factory operators and maintenance staff will test the system in a real-world environment, providing feedback on its usability, effectiveness, and performance.

3.5 UML Modeling of the Proposed System

3.5.1 Use Case Diagram

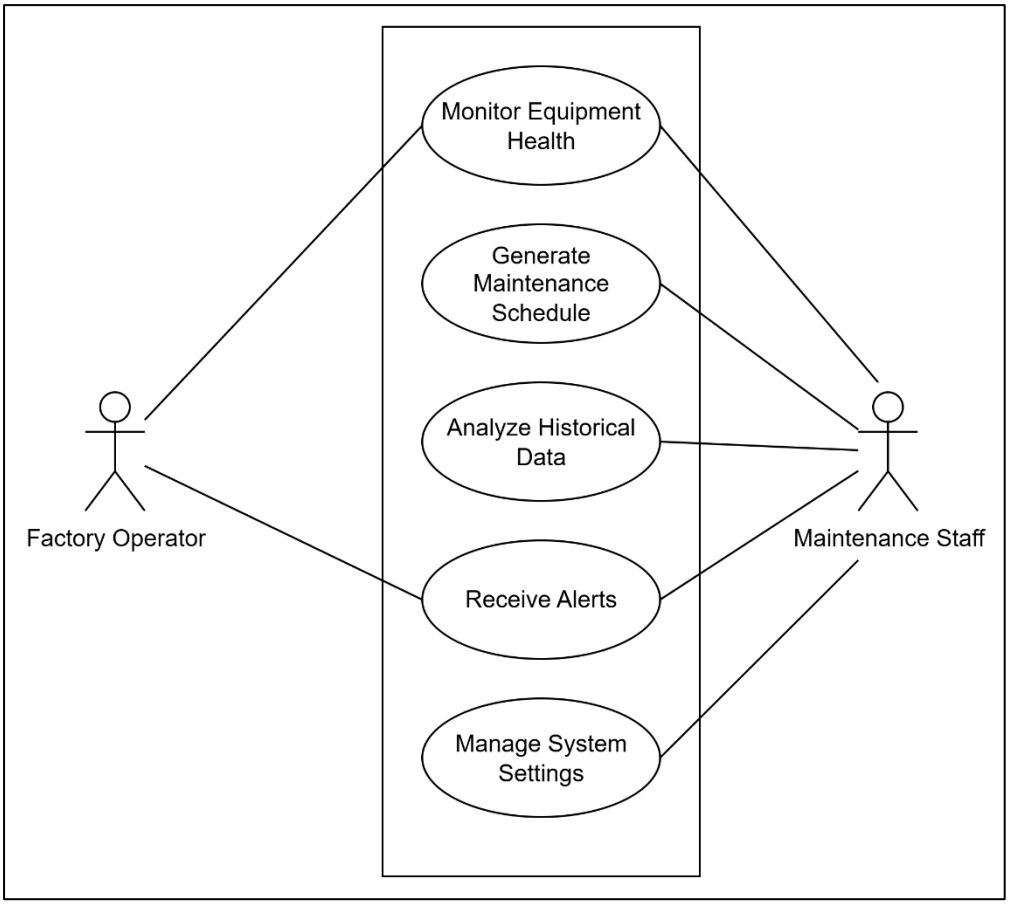


Figure 3.11: Use case Diagram for PdM System

The Use Case Diagram for the Smart Predictive Maintenance System depicts the three core relationships with the system which include Maintenance Staff and Factory Operator. Factory Operators utilize the system for operational oversight. Operators are able to monitor the status of the machines on the dashboard and receive alerts which are notifications triggered when the system detects anomalous and predicted failure conditions. These activities support the operational tasks of the production employees by providing real-time visibility of the equipment health.

The Maintenance Staff also have a system interaction role which is broader. Besides having the capability to monitor assets and receive alerts, these staff members are also responsible for the use cases of “Analyze Historical Data”, “Generate Maintenance Schedule”, and “Manage System Settings”. By analyzing historical telemetry and maintenance documentation, the maintenance team is able

to determine patterns of recurring problems which helps to fine tune predictive thresholds and confirm the machine learning model predictive accuracy. They are able to convert predictive maintenance interventions into actionable work orders using the “Generate Maintenance Schedule” use case to perform planned maintenance scheduling to perform maintenance during optimal time windows to reduce and avoid unplanned downtime. Maintenance Staff have system administrator privileges to configure notification settings, alter action parameters in the system for alerts and other system operational controls through the use case “Manage System Settings”.

In general, the revised use case diagram accurately captures the potential interactions for the developed prototype, in that all configuration, monitoring, and scheduling activities can now be performed directly by personnel within maintenance and operations, and no longer requires the involvement of a System Administrator.

3.5.2 Class Diagram

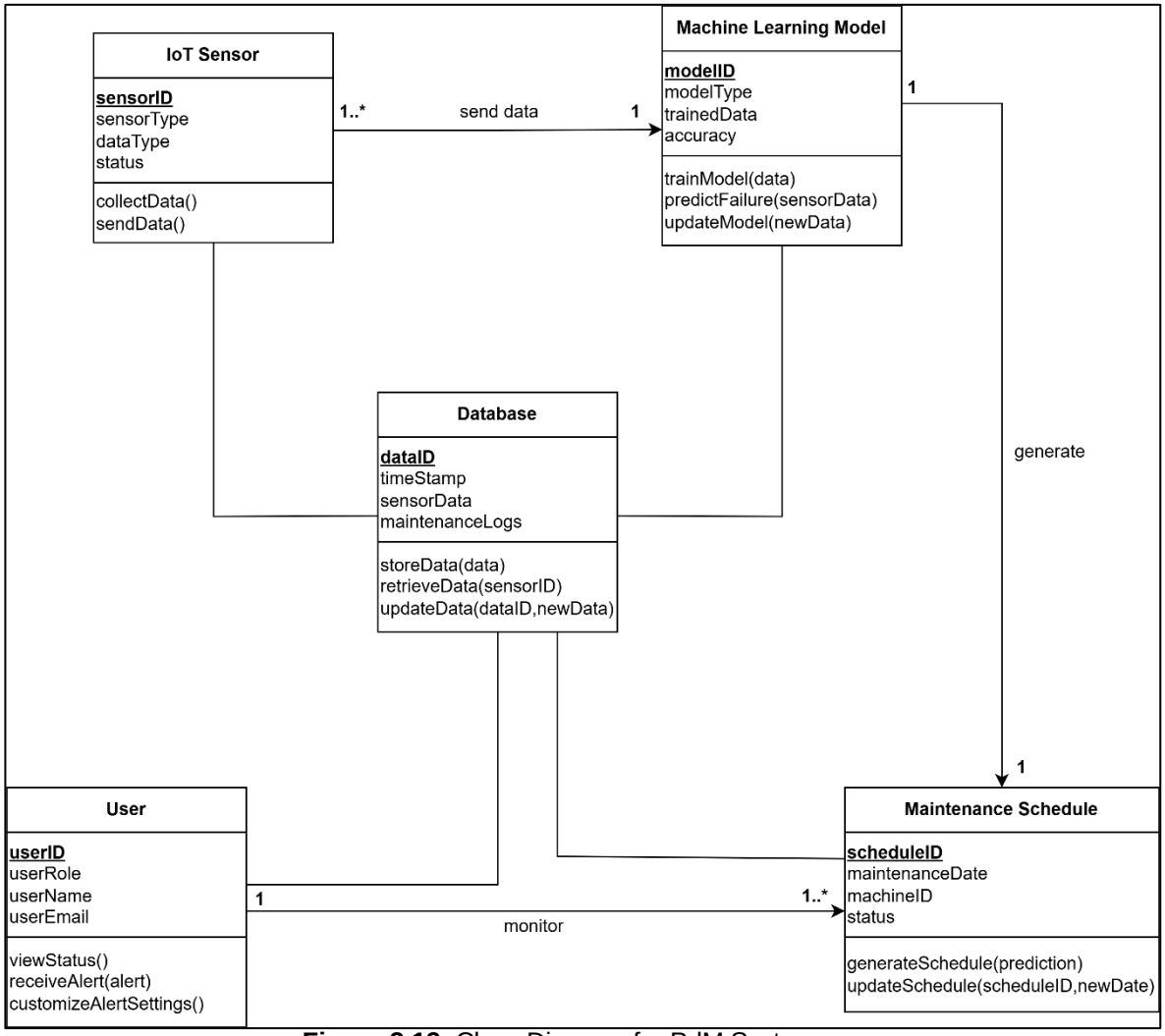


Figure 3.12: Class Diagram for PdM System

The Class Diagram provides a detailed overview of the system's structure by depicting the classes, their attributes, and methods. It defines how the different components of the predictive maintenance system interact with each other and how they are structured.

The IoT Sensor class is essential as it represents the data collection component of the system. Each sensor collects real-time data from machinery, such as temperature, pressure, and vibration. The class has attributes like sensorID, sensorType, and dataType, and methods such as collectData() and

sendData(). These sensors continuously monitor machine performance and send their data to the system for analysis.

The Machine Learning Model class forms the core of the predictive maintenance system. This class represents the algorithms that process the sensor data to predict when a machine might fail. The machine learning model has attributes such as modelID, modelType, and accuracy, and methods like trainModel() for training on historical data, and predictFailure() for making predictions based on real-time data.

The Maintenance Schedule class is used to generate and manage maintenance tasks. It stores the predicted maintenance dates and the machines that require servicing. This class ensures that maintenance is scheduled at the most optimal time to reduce downtime and prevent unexpected equipment failure. It has attributes such as scheduleID, maintenanceDate, and machineID, with methods like generateSchedule() and updateSchedule().

The User class represents the various system users (maintenance staff, operators, administrators) who interact with the system. This class includes attributes like userID, userRole, and userName, and methods such as viewStatus(), receiveAlert(), and customizeAlertSettings().

Finally, the Database class stores all system data, including sensor data, maintenance logs, and machine learning predictions. It plays a central role in ensuring data persistence and access. It has attributes like dataID, timestamp, and sensorData, and methods such as storeData() and retrieveData() to manage the system's database.

The relationships between the classes are depicted in the diagram, where the IoT Sensor sends data to the Machine Learning Model for analysis. The Machine Learning

Model generates failure predictions, which are then used to generate maintenance schedules in the Maintenance Schedule class. The User interacts with the system to view data, receive alerts, and manage maintenance tasks, while the Database ensures that all data is securely stored and accessible for future analysis.

3.5.3 Sequence Diagram

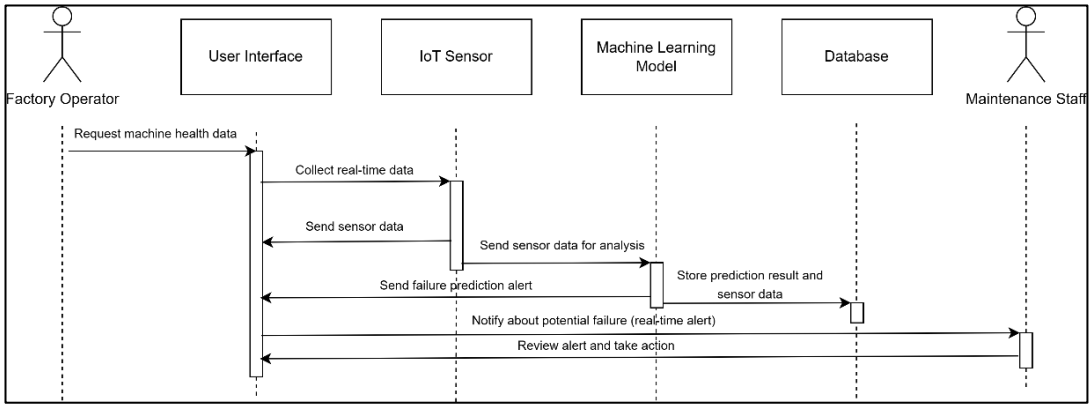


Figure 3.13: Sequence Diagram of PdM System

Example Scenario: Predicting Equipment Failure.The Predictive Maintenance

System begins with the Factory Operator, who interacts with the User Interface (UI) to request real-time machine health data. This request triggers the UI to gather data from the IoT sensors attached to the machinery. These sensors continuously monitor critical parameters such as temperature, vibration, pressure, and operating speed. The collected data is then sent to the UI, where it is displayed for the operator to assess the current state of the equipment. Simultaneously, the data is transmitted to the Machine Learning Model for further analysis, where the system begins to process the data to determine if there are any signs of potential failures.

Once the Machine Learning Model receives the sensor data, it applies predictive algorithms based on historical data patterns. The model analyzes the incoming data to detect anomalies or patterns that could indicate impending failure. If

the model identifies a potential risk, it generates a failure prediction alert, which is immediately sent to the UI and shown to the Maintenance Staff. Along with the prediction, the system logs all the sensor data and the model's predictions into the Database. This not only keeps a historical record for analysis but also allows the system to continuously learn and refine its predictive accuracy over time, improving its performance as it collects more data.

Upon receiving the alert, the Maintenance Staff can review the predictive failure warning on their UI dashboard. The staff can then decide whether to schedule maintenance, take immediate corrective action, or further investigate the issue based on the severity of the alert. By using the predictive maintenance data, the system ensures that maintenance activities are proactively scheduled before failures occur, minimizing unplanned downtime and reducing maintenance costs. The stored data in the Database is also critical for analyzing long-term trends in machine performance, helping operators make data-driven decisions to improve efficiency and extend the lifespan of machinery.

3.5.4 Activity Diagram

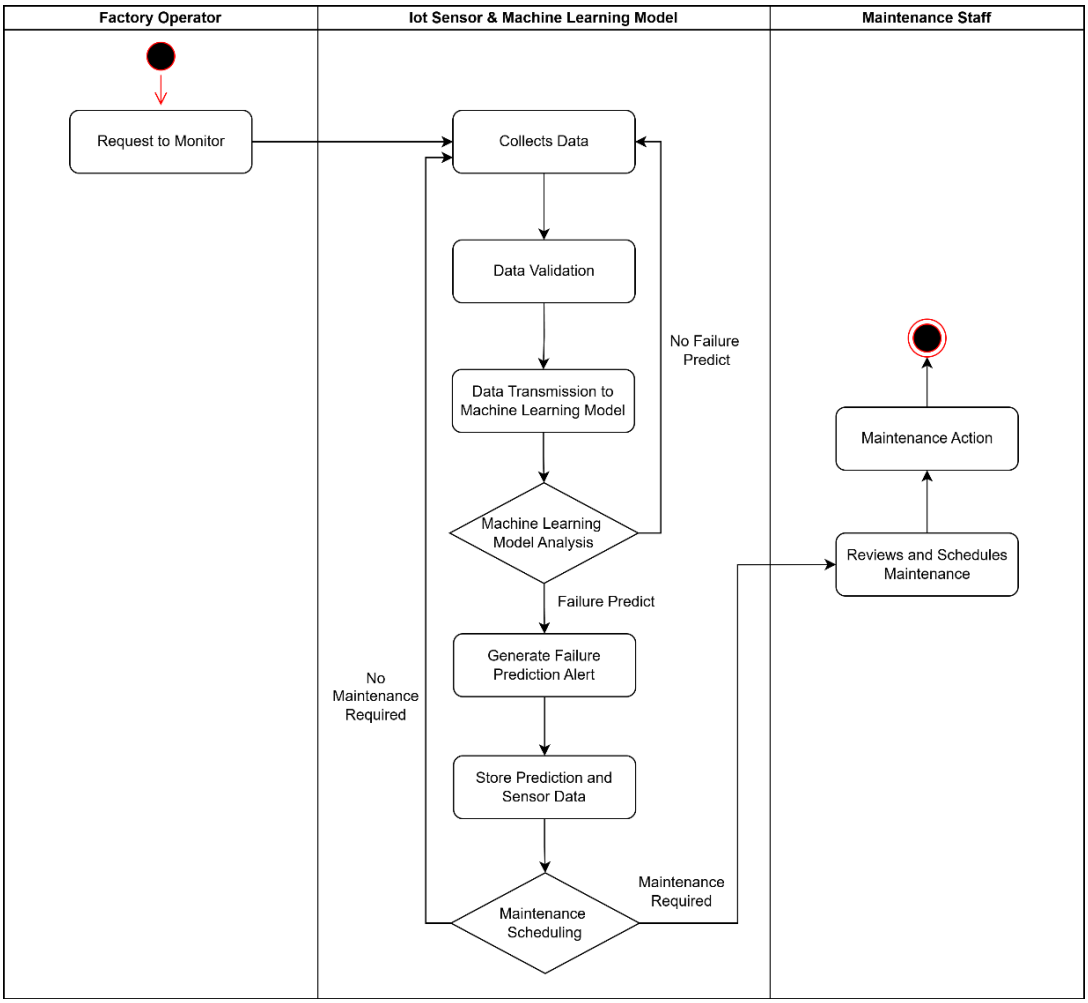


Figure 3.14: Activity Diagram for PdM System

The Predictive Maintenance System begins when the Factory Operator requests machine health data via the User Interface (UI). The system collects real-time data from the IoT sensors attached to the machinery, which is validated and displayed on the UI. If the data is valid, it is sent to the Machine Learning Model for analysis. The model processes the data to predict potential equipment failures, comparing it against historical data to identify patterns. If a failure is predicted, an alert is generated and sent to both the Factory Operator and Maintenance Staff.

Upon receiving the alert, the Maintenance Staff reviews the predicted failure and decides whether maintenance is required. If maintenance is necessary, the system automatically schedules it based on the prediction, and the maintenance logs are updated in the Database. Once maintenance is performed,

the system returns to its monitoring state, ready to handle future predictions. This flow ensures the system minimizes unplanned downtime and maintains operational efficiency through proactive maintenance scheduling.

CHAPTER 4

FINDINGS AND TESTINGS

4.1 Introduction

Chapter 4 deals with the evaluation activities designed to assess the performance, dependability, and efficiency of the Smart Predictive Maintenance System constructed for this project. This chapter, like the others, follows the constructed logic of the methodology chapter to systemically evaluate the system's outputs, validation of the machine learning model, and the interaction of the user interface, the database, and the system's real-time control and monitoring. As the system is designed for manufacturing, specifically, for small and medium-sized enterprises (SMEs), its performance for the manufacturing ecosystem of the SMEs is to be evaluated on several criteria, including the system's ability to predict, the system's responsiveness, ease of use, and performance stability in various simulated exercises.

The chapter deals with the learning generated in the course of model development and the system to be implemented to demonstrate how the machine learning algorithms used the telemetry data to determine various levels of, and the complete data sets, for equipment failure to predict, in the system's installed data set. The main focus of the study is to demonstrate the value from the machine learning to the SMEs and if the system incorporated the predicted values the stakeholders raised during the requirement and design phases.

Next, this chapter describes the types of testing that took place according to the V-Model's verification and validation phases, and include module-level unit testing, integration testing of components of the system, system testing and verification of end-to-end data flow, followed by user acceptance testing. Each test is performed to ensure that the system achieves its user-identified goals at the operational level and supports the factory operators and maintenance personnel to

accomplish their jobs.

In this chapter, the implementation of the database is also discussed, how the data telemetry, predictions, logs, and user actions are saved and how the data are retrieved and manipulated. The structure of the database and its efficiency determine how quickly and reliably the system performs real-time fault detection and maintenance scheduling. The chapter ends with a review of the findings from the testing and the evaluation of the gaps between what the system was intended to do and what was achieved, and the potential for further development that can be built from these findings.

In total, this chapter aims to confirm that the system indeed provides the user with accurate data and timely warning of potential maintenance actions that can be performed, and by doing so testifies that the developed solution caters for user needs within the operational manufacturing environment and within the scope of the project provided at the outset.

4.2 Analysis Finding

The results of the survey show that participants in different manufacturing functions face the same challenges in their maintenance activities, especially the concentration on their use of manual logbooks and simple instruments that don't allow real-time tracking. This means more unplanned downtimes and challenges in forecasting equipment breakdowns. There is also an interest in predictive maintenance, and in particular, participants want real-time notifications of failures, more accurate predictions of failures, and dashboards that are easy to use. This means the system being designed will meet the requirements of the industry as demonstrated by the interest in the features. All in all, the survey results show that there is a predictive maintenance system that captures the requirements of the users and can help organizations achieve their goals by improving their operational productivity.

4.2.1 Pilot Study

Data collection comprised an initial stage where the selected sample groups of manufacturing employees, such as factory staff, maintenance managers, and technical professionals, completed a pilot structured questionnaire for testing purposes. The clear and pertinent questions to the objectives of the comprehensive data collection, which are determining user needs, maintenance issues, and expectations from a predictive maintenance system, were validated through the pilot initial surveys. Participants made comments which communicated the questions to a higher, or perhaps more appropriate, level. The adjustment made which, in response to the remarks, were comments regarding question phrasing, the level of response to the scales, and the drift of the surveys, enhanced the over-all validity of the question. The pilot surveys confirmed that the respondents were in firm command of the questions involving maintenance practices, downtime frequency on tasks, current methods of tracking tasks, and desired outcome features. The pilot surveys

confirmed that the respondents were in firm command of the questions involving maintenance practices, downtime frequency on tasks, current methods of tracking tasks, and desired outcome features. The pilot surveys documented predicaments of the respondents which were issues concerning the overuse of paper, manual documenting, and the absence of maintenance real-time data visibility. Overall, the pilot study was of great importance in structuring the questionnaire.

4.2.2 Result and Findings of Pilot Study

Based on responses from 20 different participants across different levels in manufacturing settings, the pilot study yielded useful information and insights. The demographics of the participants show that the survey was successful in obtaining responses from a variety of respondents, and the detailed contact information further verified that participants really did complete the survey. The distribution of the of positions shows that production floor managers make up the largest proportion of the respondents, followed by maintenance managers and technical staff, which gives the researcher an additional vantage point from participants with different roles in the operational and systems level of the hierarchy. The data provided shows that respondents have a range of different years of work experience, with numerous participants reporting over 6 years in maintenance operations, which strengthens the validity of the data.

Sectors include automotive, electronics, food processing, and pharmaceuticals showing the pilot study reflects an array of manufacturing industries with disparate operational requirements. Results regarding maintenance practices indicate that the majority of establishments are still using digital-based tools and their databases are also kept at a basic level, while a considerable number still prefer using paper logs. This indicates an early stage in the adoption of predictive maintenance, and the majority of organizations have not yet included IoT sensors in their workflows. Further, the data reveals that there are unplanned downtimes with

weekly and monthly occurrences being the most predominant which illustrates the urgent need for improved equipment monitoring systems.

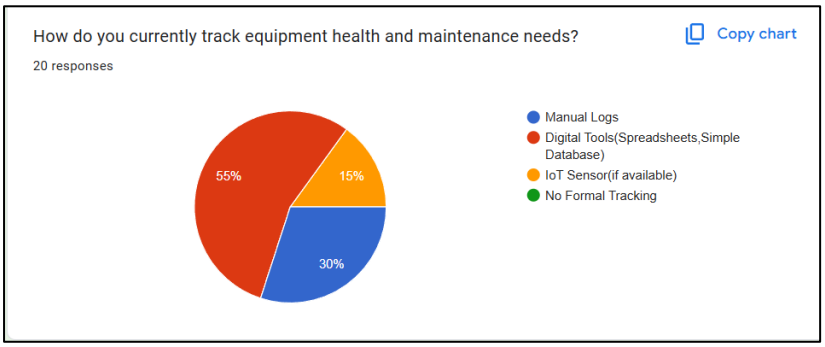


Figure 4.1: Track Equipment Health and Monitoring System

The answers indicate that more than half of the respondents are already using basic digital record-keeping tools such as spreadsheets or primitive databases. About a third of the factories are still using paper-based logbooks, and only a few have implemented IoT sensors, which indicates a very low uptake of automated record-keeping for monitoring purposes. Some respondents indicated that they have no formal tracking system, demonstrating clear weaknesses in operational record-keeping.

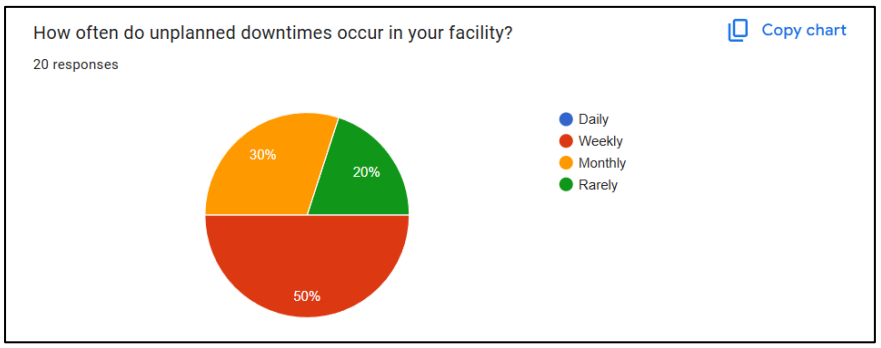


Figure 4.2: How Often Unplanned Downtime

Half of the respondents report that the most significant issue that unplanned weekly downtime is causing is the continued operation of unplanned downtime. The respondents also indicate that there are significant unplanned downtimes occurring on a monthly basis. A handful of respondents report that there are unplanned

downtimes occurring on a yearly basis, which suggests that the majority of facilities experience unplanned operational downtimes on a regular basis.

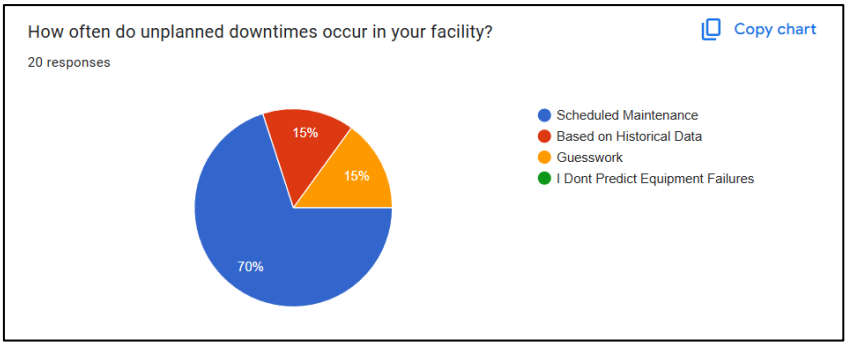


Figure 4.3: How Often Unplanned Downtime

Most of those surveyed depend exclusively on time-based maintenance, indicating a reliance on maintenance servicing, as opposed to predictive servicing. Some of them use historical data, while others acknowledge using no data as predictive guesswork to imply a lack of basic predictive frameworks. Some others completely lack predictive capabilities, which illustrates a significant gap in predictive systems.

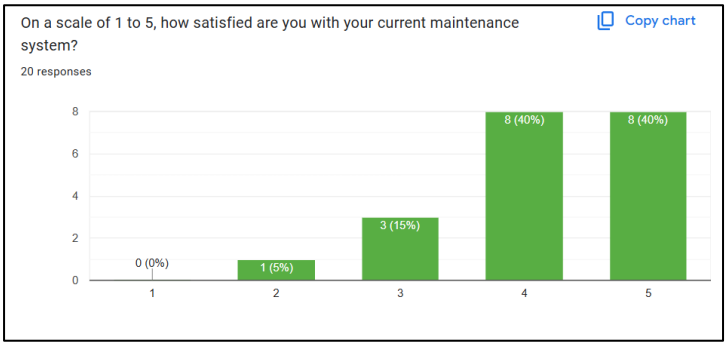


Figure 4.4: Statisfied of Current Maintenance System

A 4 or 5 was rated as general contentment with room for improvement. Slightly less than the majority were rated as mid-range satisfaction, which indicated existing systems as performing at a moderate level. There was almost no reported dissatisfaction. Thus, it can be concluded that systems in place, although functional, are likely not for ongoing systems.

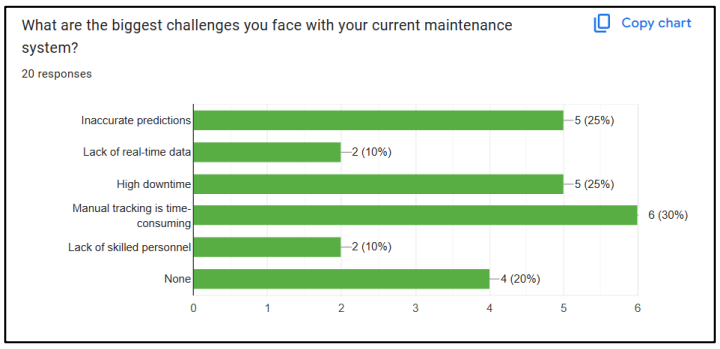


Figure 4.5: Biggest Challenges with Current Maintenance System

The problems that were reported the most were the high inefficiencies surrounding system downtime, time-consuming manual tracking, and inaccuracy within the metrics. There were some respondents who mentioned the lack of skilled personnel and a real-time data stream that shows deficiencies in the technology and workforce. There were a few who stated that there were no problems, indicating that there is a varying degree of system maturity across the different facilities.

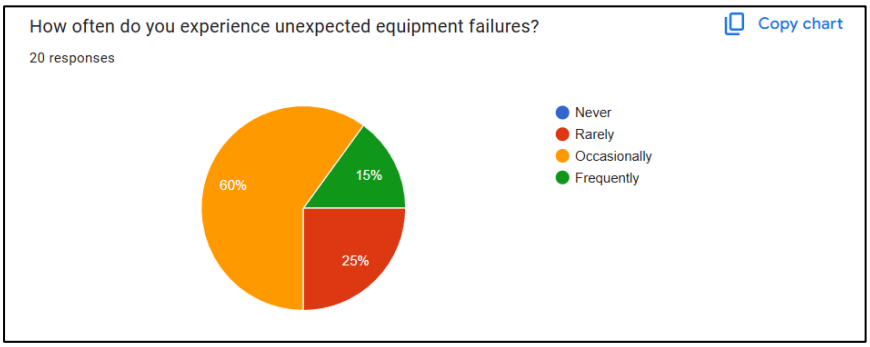


Figure 4.6: Experience Unexpected Equipment Failures

Most people experience failures from time to time, indicating that breakdowns are still occurring more often than at the bottom. Those who experience fewer failures are a smaller group. The opposite group experiences failures. This indicates that the maintenance operations are not performing the same across different locations.

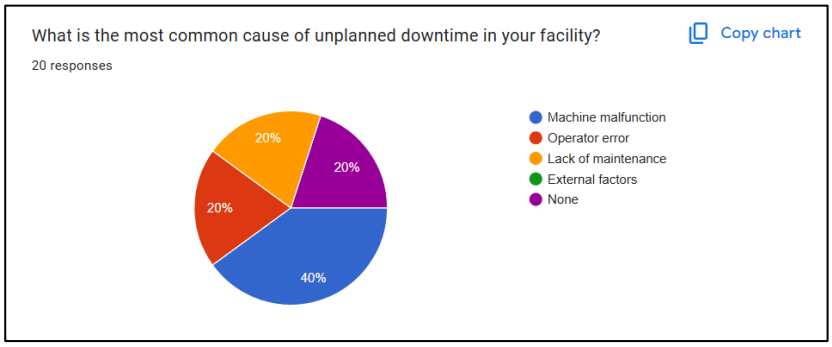


Figure 4.7: Common Cause of Unplanned Downtime

The dominant answer indicating severely deficient response reliability was the mechanical malfunction of the machine under consideration, constituting almost 50% of the responses. Further issues regarding mechanical reliability stemmed from operator errors also noted as human gaps as well as the missing or deficient maintenance of the machine. Furthermore, a few respondents noted other external factors or none at all indicating the response variability in these types of questions across numerous industries.

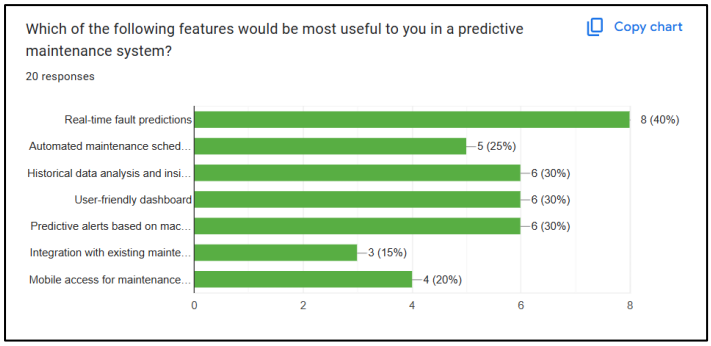


Figure 4.8: Features that Useful in Maintenance System

Real-time fault prediction was also selected as the most valuable feature by most respondents, signalling the desire for proactive detection of issues. Predictive alerts also demonstrated the desire for accessible analytical tools, which were further illustrated by the popularity of easy-to-use dashboards and historical data insights. Although additional integration with existing systems and mobile accessibility were mentioned, they were considered secondary.

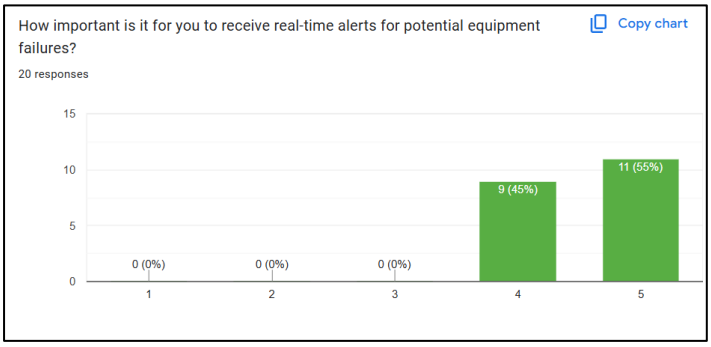


Figure 4.9: Important to Receive Real Time Alerts

All respondents rated real-time alerts as important or very important, indicating strong demand for notifications preventing failures in real-time. The high percentage of '5 - Very Important' suggests that reducing operational risk is correlated with faster response times.

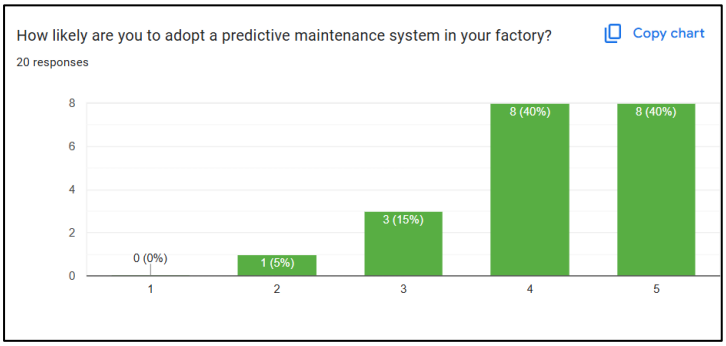


Figure 4.10: Adopt a PdM in Factory

Most participants felt their odds were high, suggesting they were very open to adopting a predictive maintenance solution if value were to be realised. There were a few respondents who selected lower ratings, indicating there is very little resistance to new technology if it is appropriately introduced.

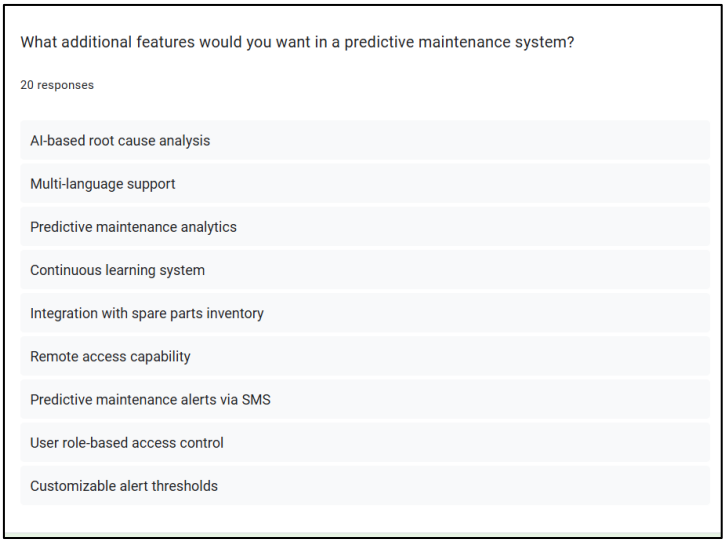


Figure 4.11: Additional Features

A large number of suggested features pointed to a distinct need for an intelligent maintenance solution, including the ability to perform AI-based root cause analysis, language translation, predictive analytics, remote access, integration with spare parts inventory, continuous learning, path deviation, and remote end-point control. From these features suggested, it could be deduced that system users are looking for an intelligent system that can adapt to the needs of the users while aiding in complex operational decisions outside of mere fault/issue detection.

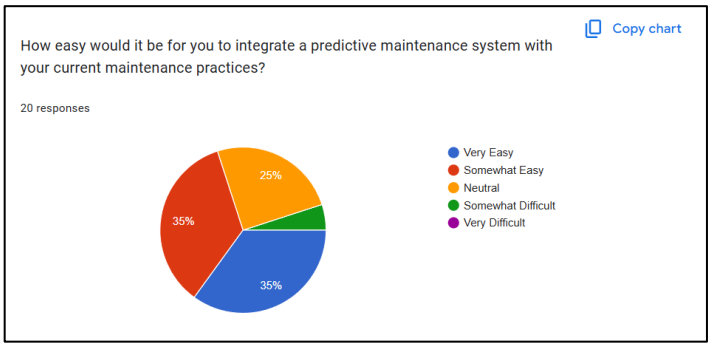


Figure 4.12: Integrate a PdM

The answers reflect an overall but somewhat divided enthusiasm about entering the integration stage of the system. Many people who filled the surveys answered that they found the integration of the system into their facilities to be “Somewhat Easy” and “Very Easy”, which is indicative of a great number of facilities already having the

framework needed to incorporate the systems. However, given that there were also “Neutral” and “Somewhat Difficult” answers, this suggests that some of the organisations who took the survey may continue to need some form of assistance or changes to their processes during or after the integration.

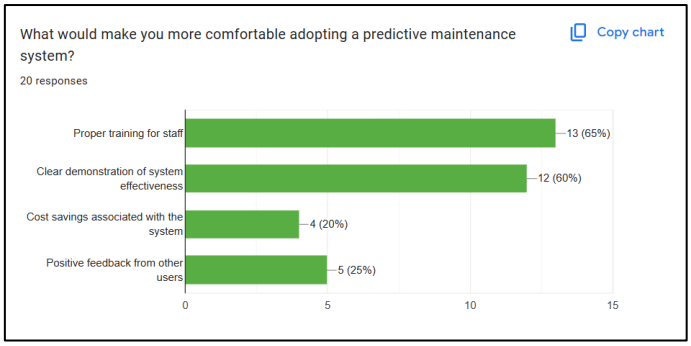


Figure 4.13: Comfortable Adopting a PdM

Participants noted that system effectiveness training and proper staff training were among the highest-rated aspects of system confidence adoption. Although cost savings are greatly valued, cost savings ranked lower, indicating that users value system clarity, ease of use, and assurance of system effectiveness more than cost savings. Additionally, indicating social proof, other participants' positive feedback was noted.

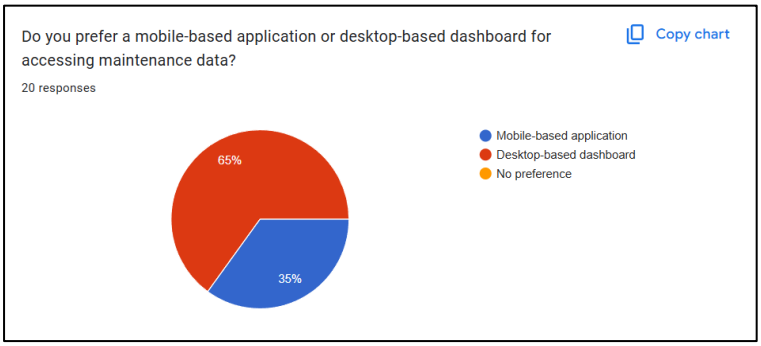


Figure 4.14: Mobile App or Desktop Website

Most respondents showed a preference for desktop dashboards, which indicated their leaning towards the more stable, large-screen control room or office dashboards. Some respondents still leaning towards the mobile dashboards demonstrated the need for flexible, on-the-go access, which justifies the requirement for mobile dashboards. However, overall, results suggest that desktop access to

dashboards is the primary operational preference.

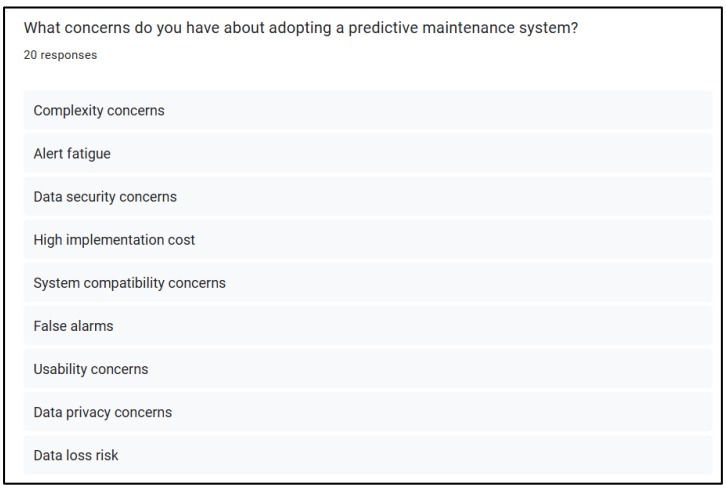


Figure 4.15: Concern About PdM

Concerns included complications associated with implementation, the potential loss of sensitive information, and the overriding expenses associated with the implementation of the system. Employee respondents noted making false predictions, lack of integration, and/or difficulty of attempts to use systems. Issues identified in the findings indicated the growing implementation of predictability-associated maintenance. Flexible, user-friendly systems to predict machine balances are expected preventative maintenance to predict balances in system usage and maintenance.

4.3 Testing and Findings

Testing and validation were done to confirm that the Smart Predictive Maintenance System was behaving as expected and that all criteria, both functional and non-functional, were satisfied, and to confirm the system worked as expected in the simulated manufacturing environment. Using the V-Model development methodology, each system component was evaluated, and accuracy of the data inputs, ML predictions, interface interactions, and database transactions were verified as validation for each component. This testing phase was performed to determine the system's ability to perform, predict, provide stability, and engage within the expected real-time parameters.

The accuracy of the machine learning model at predicting equipment failure and recognizing anomalies was evaluated using both historical and simulated data sets. The model was verified to achieve the predicted range of accuracy, precision, recall, and time. While this was occurring, validation was also occurring on the data flow pipeline to confirm the IoT data was ingested, pre-processed, predictions were generated, alerts were sent, and there was no noticeable latency.

System analysts made sure user interface testing established that maintenance personnel and factory operators could seamlessly navigate the dashboard, understand the output from the systems, and take action on real-time alerts. This validated usability goals for the system with respect to non-technical users. The system's database was also assessed to make sure there was no performance dip due to continuous data logging, maintained records of predictions, and queried historical data.

In conclusion, the results support that the system can pull and provide timely predictive information, send alerts of failures with a high degree of accuracy, and run without interruption on multiple testing scenarios, confirming the system can be deployed in SME manufacturing.

4.3.1 System Interface

There were simple benefits available to operators and maintenance workers considering the system's interface. Tests showed effortless navigation and accuracy for timely data. Dashboards retrievable in real-time showed machine health parameters and lack of lags for alerts to fail. Predictive failures were described to users with words and for gaps in the service. Historical charts captured data in an efficient range of dates. The users could pull up completed records with the maintenance scheduling predictive system in a timely service. Overall the testing was described as an interface in maintenance operations for the users.

1. Login Page

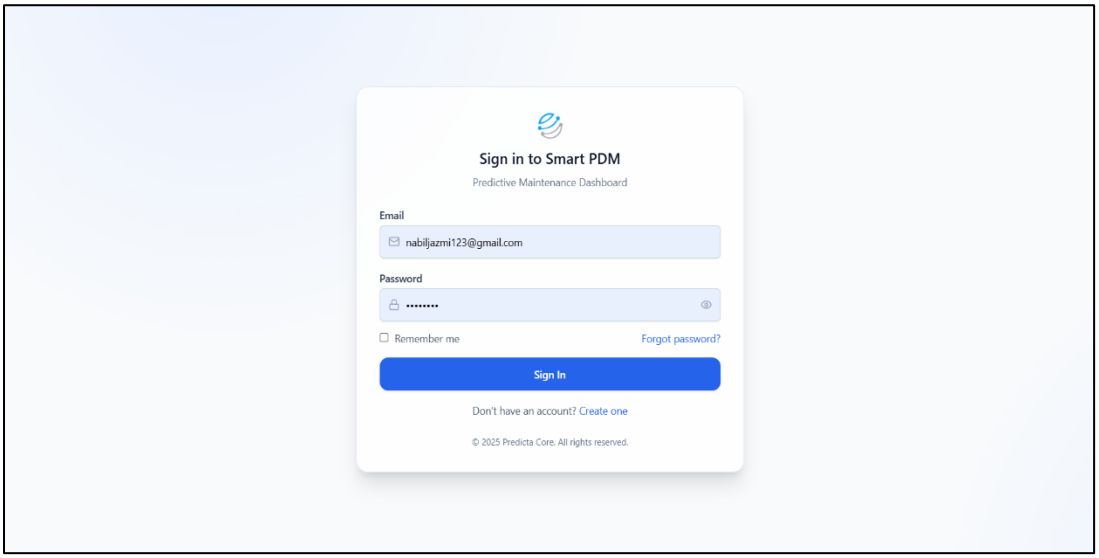


Figure 4.16: Login Page

To get to the Smart PDM system users must first complete the authentication process. Users are required to input the email and password. On the users Smart PDM system dashboard, first page they will see is the the Smart PDM system dashboard. The login page has have a low information density and is easily able to be comprehended, containing text fields, show password toggle, a remember checkbox, and quick access buttons to create and reset passwords. All data is maintanined and protected from operators, maintenance and system managers.

2. Signup Page

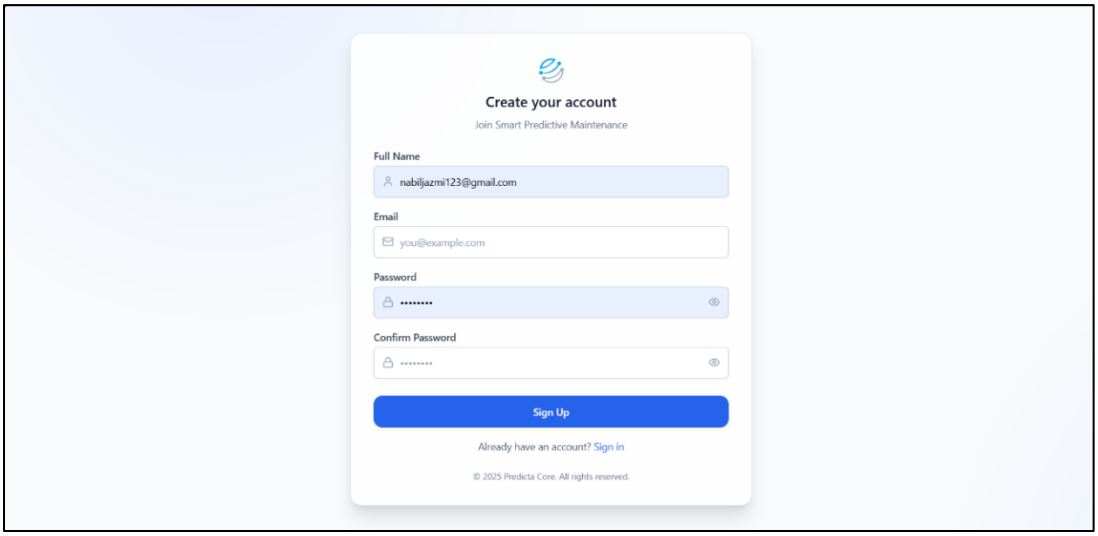


Figure 4.17: Signup Page

The Signup Page makes it easy for newcomers to the Smart Predictive Maintenance system to create new accounts. New users fill out their full name, email address, password, password confirmation to ensure account security, as well as to help avoid erroneous inputs. New users will find their initial registration to be simple, in part due to the uncluttered interface and well-labeled fields. Existing users will find convenience in the system's seamless design, which will allow them to easily find the login page.

3. Dashboard Page

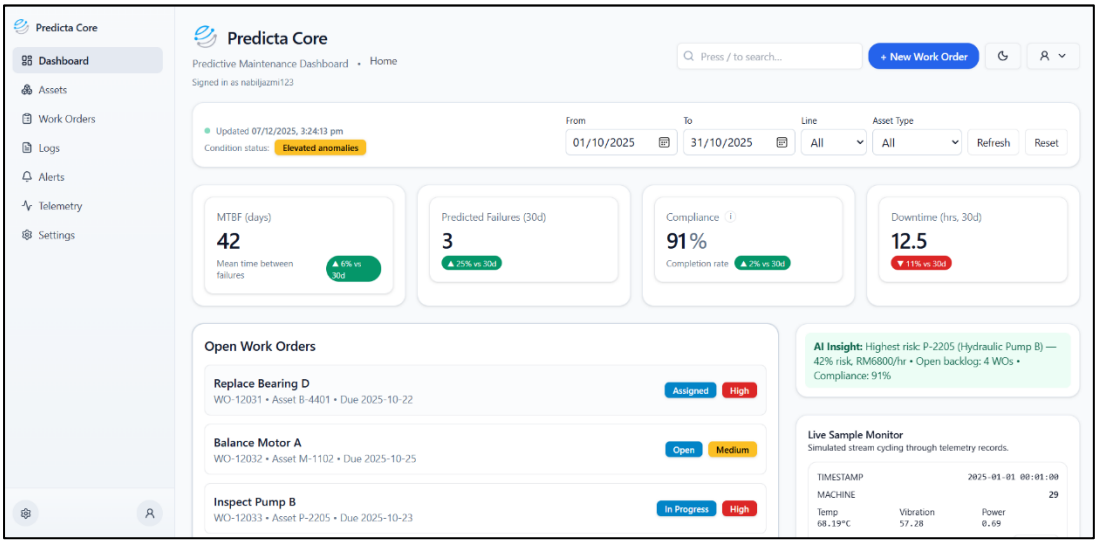


Figure 4.18: Upper Dashboard Page

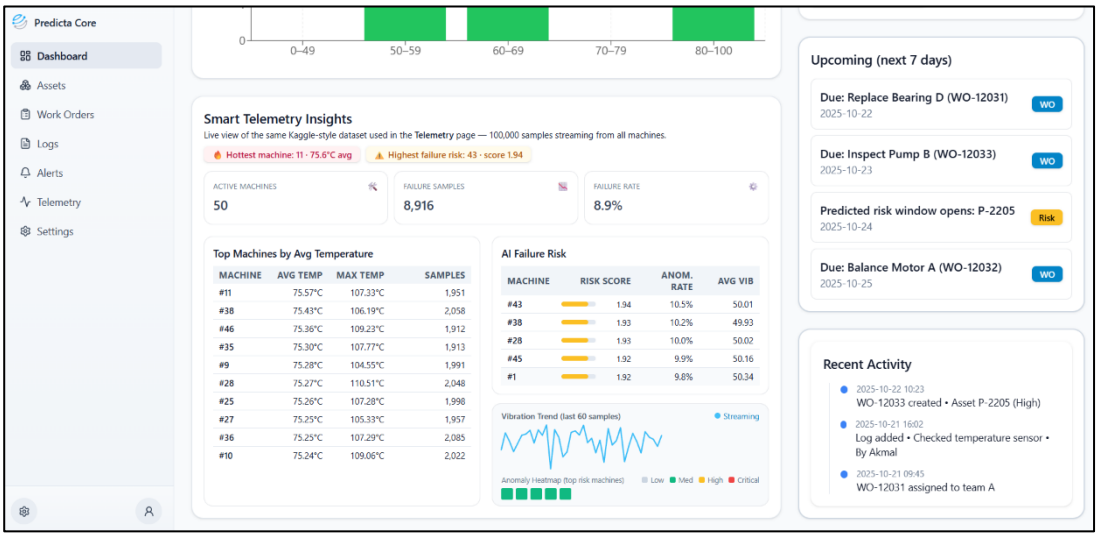


Figure 4.19: Bottom Dashboard Page

The Dashboard Page is the organizational center of the Smart Predictive Predictive Maintenace System. It provides a summary of the status of each machine with regards to its health, performance, and maintenance states. The MTBF (Mean Time Between Failures) , predicted failures, compliance, and total downtime metrics are always available. The top of the dashboard displays total downtime, and the other metrics for a summiary overview of the operational status. The interface displays all active work orders, and the AI insight for major high-risk assets. Engineetd telemetry with temper g4ad, and active power is also encapsuled in a subsrface. Finally, the dashboard displays all maintenance tasks that are due within a calendar period

along with a short summary of user activities. With these, the user is capable of reviewing the maintenance calendar alongside system activities without switching screens. The user friendly design of the dashboard is centered around the fundamental goal of enabling the user to quickly and easily access high quality data of operational relevance.

4. Assets Page

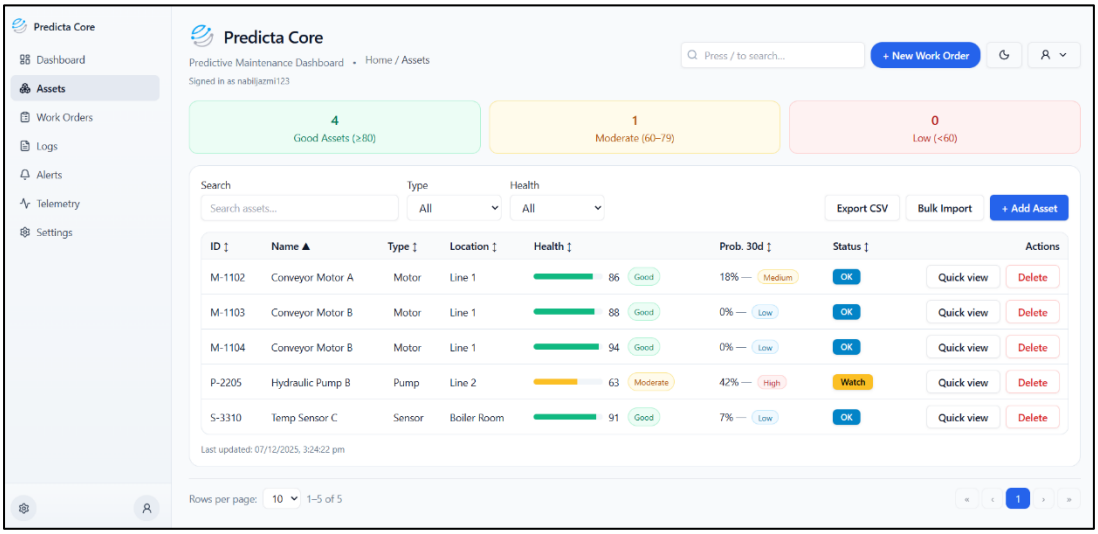


Figure 4.20: Assets Page

The Assets Page contains a list of all equipment registered in the Smart Predictive Maintenance System. Users can view and monitor the health and risk of the equipment and see the operational state of the pieces of equipment. Each of the listed assets contains the following attributes: id, name, type, location, current health score, 30-day probability of failure, operational status, and health score. To enhance the user experience and allow users to quickly monitor the operational state of the assets, they have integrated a color health score value based on the following categories: Good, Moderate, Low Risk, High Risk, Watch. Users have the flexibility to customize their asset view based on their location, type, operational status, and health status using the built-in search and filter functionality. Users can add, bulk import, and export assets in CSV format, secondary assets and asset details in quick view, and secondary asset details in detail view which helps maint managers manage maint interfeces so that they can manage equipment conditions to maint

interfeces so that maint managers can efficiently manage equipment condition and prioritize maint interfeces to manage maint equipment condition and maint interfeces so that maint managers can efficiently manage equipment condition and prioritize maint planning.

5. Work Orders Page

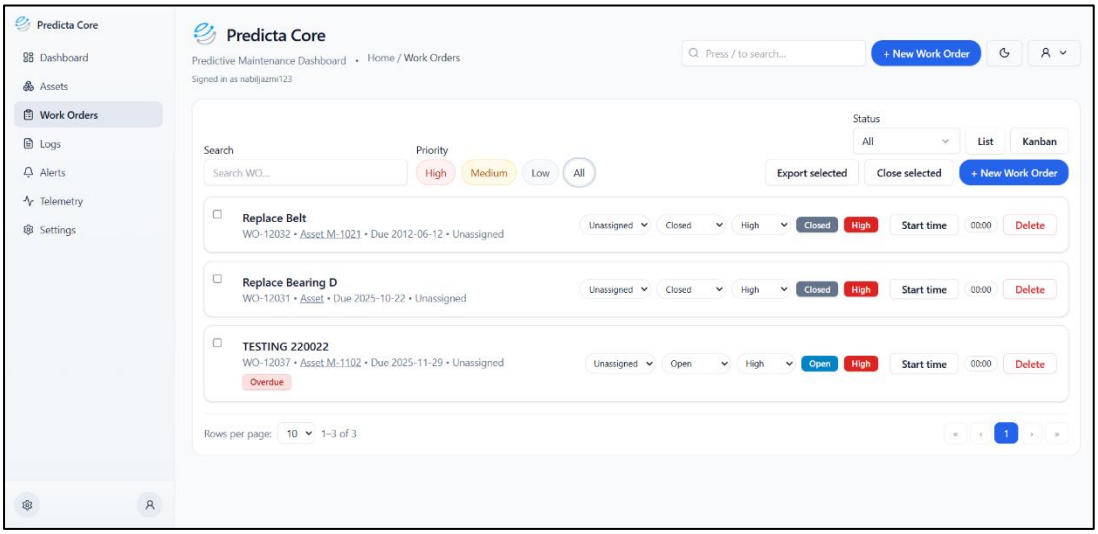


Figure 4.21: Work Orders Page

In the system, the Work Orders page gives users visibility on all maintenance-related tasks in the system in an organized fashion so users can track and manage work. Each work order has the asset name, date due, prioritization, and work order status so maintenance staff can see and identify overdue work and what tasks are overdue. Users are able to filter work orders by prioritization and work order status to ease navigation. Users can manage work order tasks and ongoing activities using Start time, Close selected, Export selected, and New Work Order. This system enables ongoing maintenance work to remain organized.

6. Logs Page

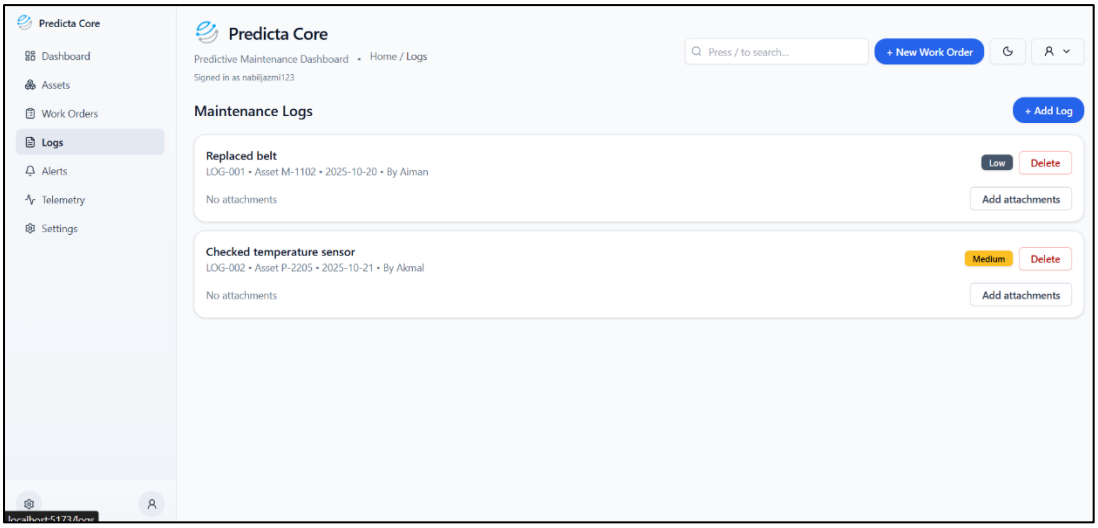


Figure 4.22: Logs Page

The Logs Page contains a centralized repository of all records pertaining to activities of maintenance to permit users to follow all actions performed on every asset. Each of the entries of the log displays the log ID, the asset that is tied to the entry, the date of the activity, and the staff member who is responsible for the activity. The display also shows the levels of priority (low, medium, high) to enable the users to focus on particular events of greater importance. Users also have the ability to delete logs and to augment them with attachments to permit the log to contain relevant documentation. This is the reason that the page serves to provide the custodians of maintenance an accurate history of the maintenance and is able to allow traceability within the predictive maintenance.

7. Alerts Page

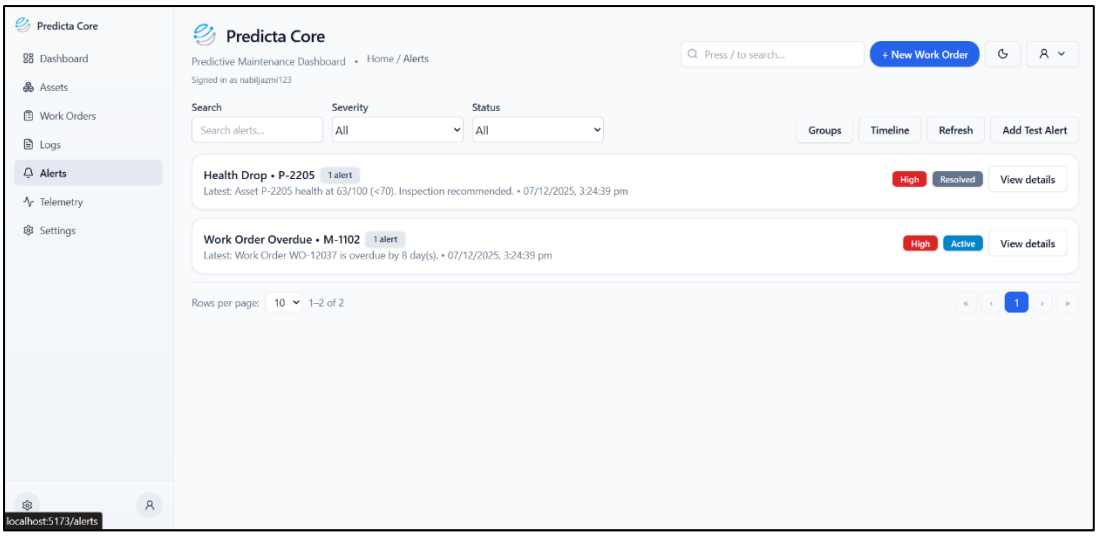


Figure 4.23: Alerts Page

The Alerts page serves as a repository for all system (auto) generated alerts detailing the base equipment conditions and maintenance notifications for all alerts by equipment condition and maintenance activity. Each alert is tied to an asset value and crosses to an impact severity value with an alert status and time stamp to allow users to focus on urgent matters first, for example, health drops or overdue work orders. Filtering by severity and status is available to maintenance teams to allow for focus on all opened alerts and all closed alerts. The users analyze the alerts to determine the appropriate action or the need to escalate. These functions make the alerts page valuable for all users requiring maintenance monitoring.

8. Telemetry Page



Figure 4.24: Upper Telemetry Pages

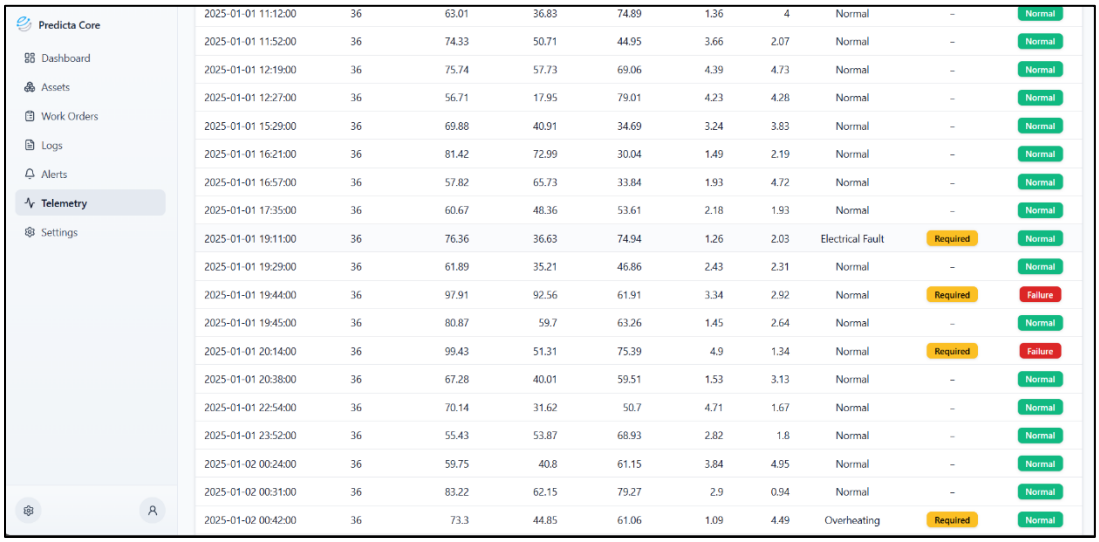


Figure 4.25: Bottom Telemetry Pages

Clients can monitor on a real-time basis the performance of a machine through a collection of 100,000 performance samples streamed and recorded from diverse machines. The telemetry page provides a wide range of performance data visualization. From the telemetry page, users can choose a specific machines and visualize the performance data in the form of dynamic graphs. Engine performance temperature, vibration, and humidity patterns and possible failures can be discovered and monitored with the dynamic graphs. The telemetry records under the dynamic graphs include the time of sample recording, temperatures and humidity

sensor data, operating mode, maintenance mode, and machine status (normal or failure). This page offers engineers data required for effectively and accurately predicting performance failures.

9. Settings Page

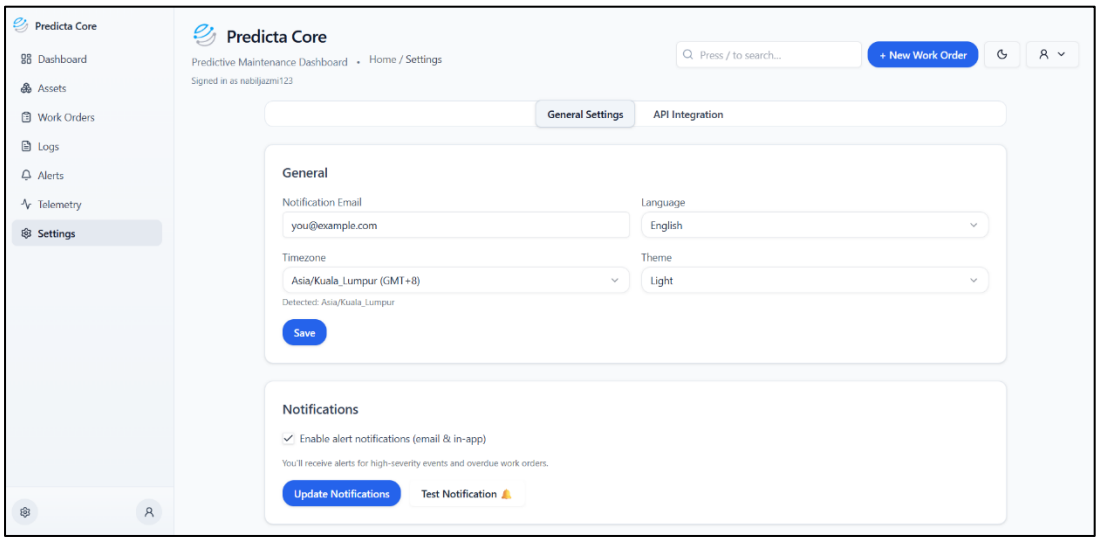


Figure 4.26: Settings Page

Users are given control over some personalization and system configuration settings within the Predicta Core dashboard. On this page, users can update their email for notifications, select their preferred language, establish the system timezone, and toggle between light or dark mode. There are also notification management settings that allow users to activate or deactivate notifications for alerts of high severity and to take a test of the notification system to verify that it is functioning correctly. This page helps to ensure that the system can be modified to address individual preferences and operational requirements, creating better usability and user experience.

10. User Profile Pages

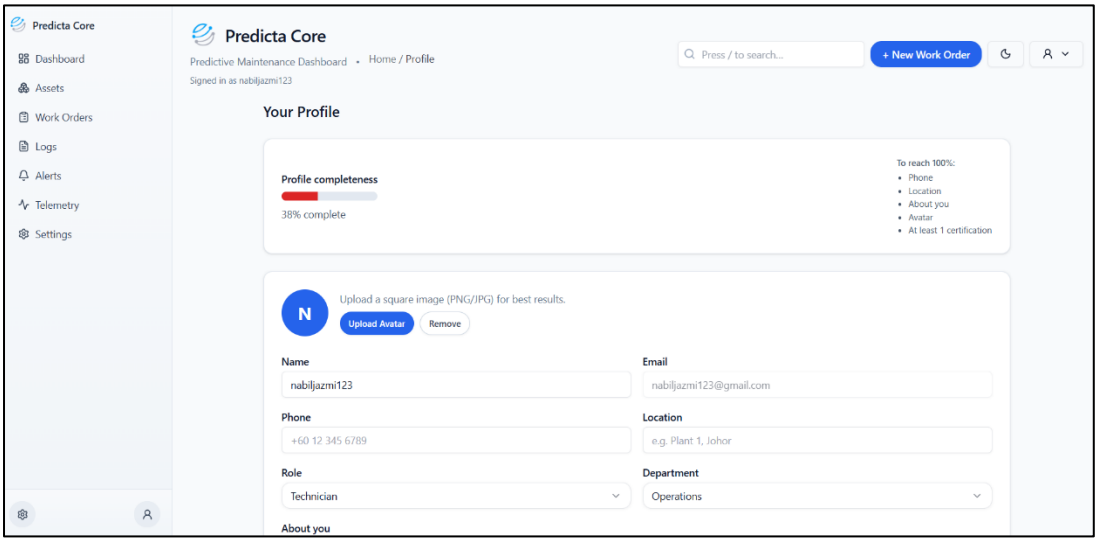


Figure 4.27: User Profile Pages

The User Profile page enables users to manage and edit their personal data within the Predicta Core system. There is a profile completeness indicator and entry fields for first and last name, phone number, address, position, department, and a brief description about the user. Users may also customize their profile by adding and deleting an avatar. This page ensures user information is accurately maintained and updated to enhance system customization, access control by role, and communication efficiency in the maintenance management process.

4.4 Database Design and Implementation

The data base design for the Smart Predictive Maintenance System is critical to the efficient functioning of the system, the proper handling of data, and the system's dependability and seamless operation. Because of the heavy reliance on real-time telemetry readings, historical datasets, and workflows of maintenance, the database is modeled using a structured relational model that prioritizes and balances dependability, scalability, and integrity of the system. The model design is primarily focused on four core entities Telemetry, Predictions, Assets, and Work Orders developed to assist in the functions of the system.

Due to the telemetry's real-time readings of temperature, vibration, and power usage of several machines, the telemetry table is mainly optimized for high-volume, high-frequency data ingestion. In real-time and up-to-date queries, history queries and telemetry data would be relevant for instant insight. Predictive outcomes from machine learning, specifically predictive classes in a rolling time window and confidence values, are contained in the predictions table. This structure is useful not just for the model to be validated and traced but also for performance validation over time so that future model retraining can be facilitated.

The design of the Assets and Work Orders tables was focused on maintenance operations and aimed at recording the identities of the machines, their current state of health, their maintenance histories, and what tasks they have already been assigned. In this, the Assets can be monitored by their type, location, and status so that managers can easily view and track condition trends. On the other hand, the Work Orders table deals with both anticipated and non-scheduled tasks and contains information on their priority and due date, their progress, and who was assigned the task. The design of the database included foreign keys so that referential integrity would be maintained across all entities, allowing telemetry and where applicable, predictive analyses, data to be accurately assigned to individual maintenance actions.

4.5 Testing Phase

The testing phase of the Smart Predictive Maintenance System took place at the first site of the Shah Alam Selangor Mid Tier Machining Company, Seri Murni Precision Manufacturing Sdn. Bhd. With the presence of a complete set of CNC machining lines, the environment was optimal for testing the System's industrial performance. Even though the System used telemetry data, the goal of the testing was to find the system's viability for ease of use, the clarity of the command, and the products for processing fit to meet the demand of SME Level Manufacturing.

The demonstrations focused on the machine operators and floor maintenance attendants at each workstation of the factory. The floor audience was given a first-hand look at the dashboard and control unit, and the demonstration team was able to examine live operational data with noise, motion, etc. flows of a factory environment. During the demonstration, operators were able to use the control dashboard to see the status screens of machines in operation, the failure prediction screens, and the screens which recorded, organized, and closed machine control commands in a work order. Even though the System screens were without real IoT input, telemetry data helped the operators see the potential of real data to ease their work. All participants were inside the factory and, they reported the dashboard was readable, easy to learn, and pleasant throughout the demonstration.

One review session took place within a small office within the facility. A production supervisor was able to analyse the system’s analytics modules within a quieter space. The Supervisor examined asset filters, records of maintenance, failure probability indicators, and the configuration of the historical data tables. This review indicated that users without a data analytics specialisation but possessing basic technical skills could easily make sense of the data shown on the screens. The supervisor said that a system of that nature could profoundly decrease the factory's reliance on manually-kept records and enhance the facility's scheduling of preventive maintenance.

The testing phase in general showed that the system functioned properly and without issues with the factory's standard hardware and that the predictive dashboard was functional and usable on operator and supervisor levels. The demonstration confirmed the system could be used and the collected feedback was thoughtful and would be used toward the next planned enhancements and integrations of actual sensors.



Figure 4.28: Interview with Factory Operator

4.6 Discussion and Findings of Testing

The assessment of the on-site demonstration and user engagement taking place at Seri Murni Precision Manufacturing Sdn. Bhd. provided valuable information regarding the system's relevance and applicability. Despite the simulated data utilized for the test, the hands-on assessment showcased the ease with which the Smart Predictive Maintenance System is implemented into an SME manufacturing context. One of the main outcomes is the remarkable responsiveness of the users. Operators and supervisors alike noted that the dashboard design was user-friendly, and that the machine health indicators, organized into color-coded systems, offered an understanding of the machine health status and the of the dynamic operational conditions. Their response indicates that the system was intuitive enough to limit the training requirements on the production floor.

Testing continues to demonstrate that the system is appropriately integrated with a set of processes focused on equipment maintenance. One of the maintenance technicians observed that a unified location for machine status, potential problems, and scheduled activities would alleviate the burden of manual logbook maintenance and other disparate communication methods. Further, the production supervisor explained that system access, with its early warning indicators and historical data, would be helpful to improve scheduling to minimize the opportunity for breakdowns and facilitate more proactive scheduling to avoid reactive breakdown response. These observations validate that the system is indeed capable of improving operational planning and efficiency.

Deployment of predictive maintenance systems is straightforward given the conditions of the factory. The factory is stable Wi-Fi connectivity and digital systems and laptops are in place which indicates that there are little to no alteration of the digital infrastructure. Employee feedback suggests there is support for the integration of predictive maintenance systems with real vibration and real time temperature monitoring. Evidence of strong predictive maintenance systems was the

potential for scalability in the continued operations of the SME.

The feedback sessions confirmed the smart predictive maintenance system was designed to be easy to use and integrate seamlessly with real factory operations for to support in improving decision-making around maintenance. Evidence of the integration of real time monitoring to the predictive maintenance system delivered the foundation the concept building for the next stages of development and was the result of the validation gained to the concept and the feedback provided by both operators and supervisors.

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4.7 Conclusion

To summarize, this chapter completed the verification activities and results on the smart predictive maintenance system and confirmed the success of the developed solution in meeting the smart predictive maintenance system's objectives set during the inception of the project. The completed system with the remarkable work done on the database design, modular implementation of the system, and the testing of the system in the simulation of the industrial environment has proven to adhered to the V-model system engineering approach. The system completed the simulation in the testing of the system in the industrial environment and demonstrated the significance of the system to the industrial environments as it processed the telemetry in real time and meaningfully predicted and delivered actionable recommendations.

It was also confirmed during the testing that user of the system, as maintenance operators, can be maintenance operators that have any technical background as they are able to comprehend the dashboards, the alerts, and work order and maintain the system with feedback to the UAT with encouragement on the perceived operational improvements, reduced maintenance delays, and maintenance decision making during maintenance operation.

It has been established that the smart predictive maintenance system is technically complete, user friendly predictive maintenance solution, that logically supports SME manufacturers to transition to the predictive maintenance model, and thus offers system a platform for the incorporation of other devices, the setting of the predictive maintenance model on automated processes, and the more wide-spread use of the system as a mobile system in order for the solution to extend the system usefulness and sustain the demand for the solution in the future.

CHAPTER 5

SUMMARY

5.1 Summary of the Research

The outcomes of the project focused on the creation of a Smart Predictive Maintenance System utilizing supervised machine learning, IoT telemetry, and dashboards for the greater efficiency of maintenance and reduction of the unplanned downtime for especially small and medium-sized manufacturing companies. While the research was being executed, the study pinpointed specific deficiencies within the traditional maintenance approaches of over reliance on reactive repairs, gaps in preventive maintenance and an acute absence of real-time machine metrics. The above gaps were validated through user surveys and current practices within the industries were assessed.

The challenges were addressed within the system by the implementation of a structured V-Model in an iterative fashion, wherein the requirements were validated extensively through the progression of each phase of the development. The implementation of a complete ecosystem of IoT data ingestion, real-time visualization interfaces, and databases for anomaly detection in machine learning was a success. Random Forest and SVM machine learning models were trained on simulated telemetry data that was able to classify the machine states and predict early machine failures. At the same time, UI interfaces were created to provide specialists, managers, and operators with the necessary information to allow rapid, educated decisions.

The system underwent extensive validation through unit, integration, system, and user acceptance testing to assess its accuracy, usability, and responsiveness. The system’s ability to receive telemetry data, trigger alerts, and perform seamlessly under simulated industrial load conditions was verified. Users were able to monitor and control their equipment with greater facility thanks to the

dashboards, visual cues, and predictive capabilities.

This research demonstrated that an affordable, scalable, and user-centered predictive maintenance system can significantly aid SMEs in their efforts to embrace Industry 4.0. The solution enhances the ability to operate efficiently, streamlines processes, and mitigates the adverse impacts caused by unpredictable equipment downtime.

5.1.1 Challenges of the Project

Throughout the development and evaluation phases, several technical and operational challenges emerged:

1. **Handling and Simulating Real-Time Data**

   Due to the rarity of real industrial sensor data from SMEs, the project predominantly used large-scale simulated datasets. Preprocessing and tuning were needed to ensure the synthetic samples closely matched the real industrial condition. Performance was also needed to be sustained while handling 100,000+ telemetry samples; this introduced the need to optimize data pipeline and avoid latency, which also came with challenges.

2. **Constraints on the Performance of Machine Learning Models**

   Predicting with the needed accuracy for consistency required further iterations of training, parameter tuning, and cross-validation. Overfitting and misclassification due to erroneous telemetry data were issues faced by some of the earlier models. Compounding the complexity of development was the need to ensure the models could generalize across a number of different machine types.

3. **Integration of Multiple Systems**

   Integrating the user interface, backend processing, and the database required a lot of detailed synchronization, which was further complicated by the need to maintain real-time rendering. Sometimes, trivial mismatches in data schemas

and API responses could lead to a breakdown of real time prediction rendering and display. The need to maintain seamless communication between the different modules while also preventing high latency was a constant challenge.

4. Considerations for the Usability of Those Without Technical Knowledge

Creating a sufficiently powerful dashboard for the technicians with minimal computer science background, while also not making it too complicated, was an iterative process. The need for improvements to telemetry insights layout, more visually clear indicators, and more alerting systems were the major points highlighted in the pilot study feedback.

5. Deployment Limitations

The system is currently validated within a simulation. There are potential challenges such as IoT sensor reliability, network consistency, noise interference, and legacy factory system integration. IoT sensor and network stability system integration. Despite these challenges, the project was able to produce a working and reliable predictive maintenance prototype that meets user and operational needs.

5.2 Contributions of the Project

The project makes several significant practical, academic, and technological contributions:

1. **Construction of Integrated Predictive Maintenance Solution**

   The project delivers the complete startup tools such as IoT simulation, ML prediction and scheduling and processing in real time, and dashboards. Such systems are rarely offered in the complete form in research solutions focused on SMEs.

2. **Maintenance Solutions that are Scalable and Affordable for SMEs**

   As opposed to enterprise solutions that come with expensive infrastructure, this system is lightweight, modular, and deployable with standard equipment and open source systems. It shows that even smaller manufacturing enterprises can avail of predictive maintenance.

3. **Machine Learning Models to identify Early Failure**

   The model has been trained to detect various telemetry streams for predictive maintenance anomaly patterns, prolonging breakdown maintenance and expensive downtimes to structure maintenance on data.

4. **Customer-Focused UX Dashboard**

   The system features a complete professional user interface with dashboards, telemetry alerts, telemetry mosaics, log files, asset control, configuration and user profiles. Such instruments enhance workflow, insight, and decision quality.

5. **Featuring Contribution to Industry 4.0 Adoption**

   The project incorporates digital prediction, real-time control and automation, supporting the digital transition and intelligent manufacturing ecosystems for SMEs.

6. Academic Contribution

Contributions made in this research extend to the ever-increasing knowledge on data-supported predictive maintenance strategies as this research work presents an actionable evidenced methodology, an approach for implementation, as well as a test framework to assist in future endeavors related to the use of artificial intelligence in industrial systems.

5.3 Future Works

While the current system demonstrates strong performance, there are significant opportunities for expansion and enhancement:

1. Integration with Real IoT Hardware Sensors

Future iterations should incorporate:

- Vibration sensors, temperature probes, current sensors
- Industrial communication protocols (MQTT, Modbus, OPC-UA) to validate system performance under real factory conditions.

2. Advanced Machine Learning Models

The use of deep learning (LSTM, CNN, Autoencoders) could improve anomaly detection and prognostics, especially for complex time-series waveform patterns.

Future versions may also incorporate:

- Ensemble learning
- Hybrid ML + physics-based models
- Remaining Useful Life (RUL) predictions

3. Implementation of Edge Computing

Deploying lightweight ML models at the machine level like on Raspberry Pi or ESP32 could reduce latency and improve reliability in environments with weak network connectivity.

4. **Mobile Application Development**

   Although the dashboard functions well on desktop, a dedicated Android/iOS app would allow technicians to:

   - Receive push notifications
   - Scan QR codes on machines
   - Perform offline inspections
   - Access remote dashboards

5. **Expansion into Multisite Smart Factory Networks**

   Future versions can connect multiple factories, enabling:

   - Centralized analytics
   - Cross-site benchmarking
   - Cloud-scale predictive insights

6. **Automated Root Cause Analysis (RCA)**

   Integrating explainable AI (XAI) tools such as SHAP or LIME would help operators understand why a failure is predicted, increasing trust and adoption.

7. **Integration with Inventory and Maintenance Management Systems**

   Automatically linking predicted failures to spare parts inventory and technician scheduling systems could transform the tool into a complete maintenance ecosystem.

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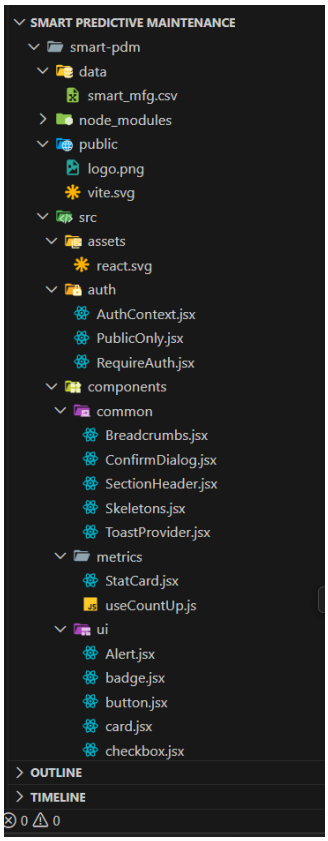
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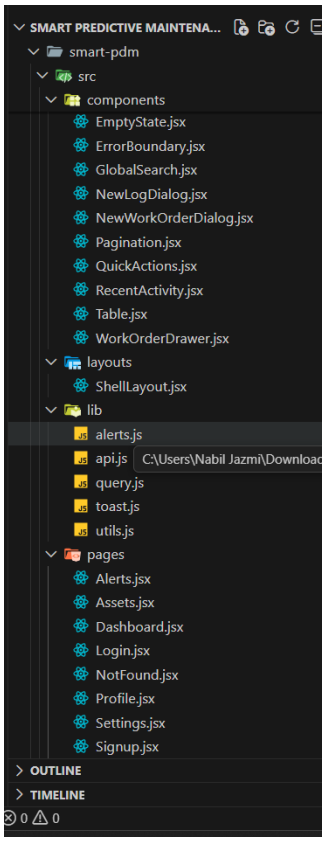
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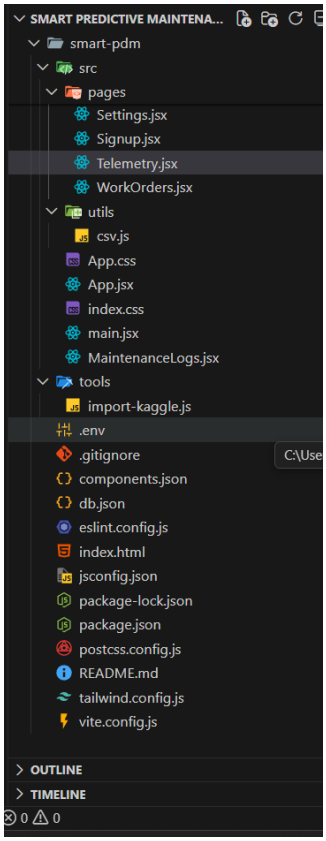
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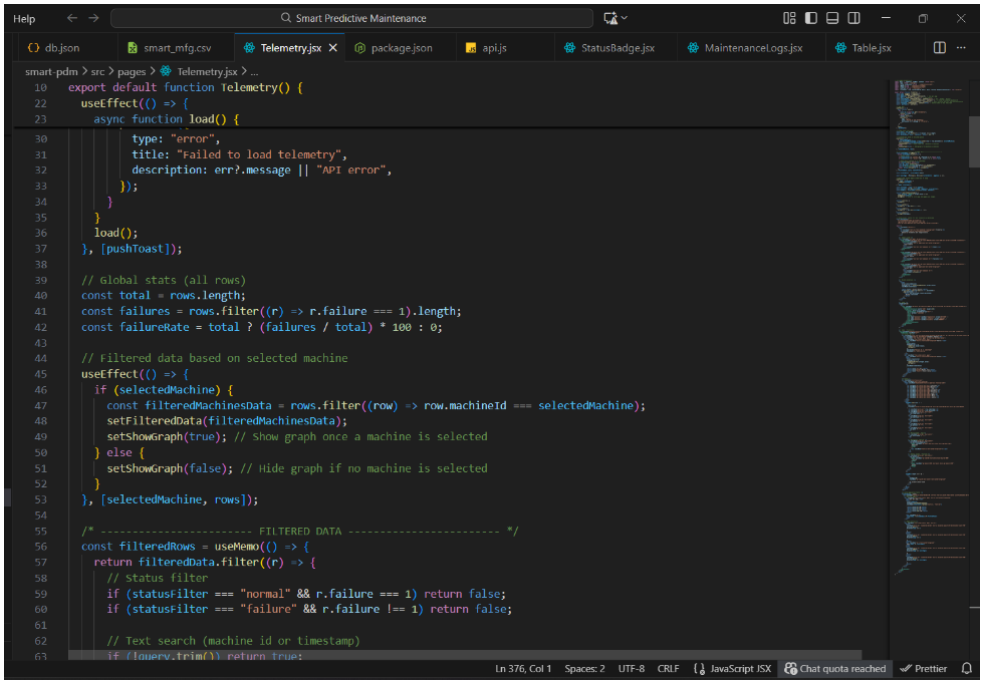
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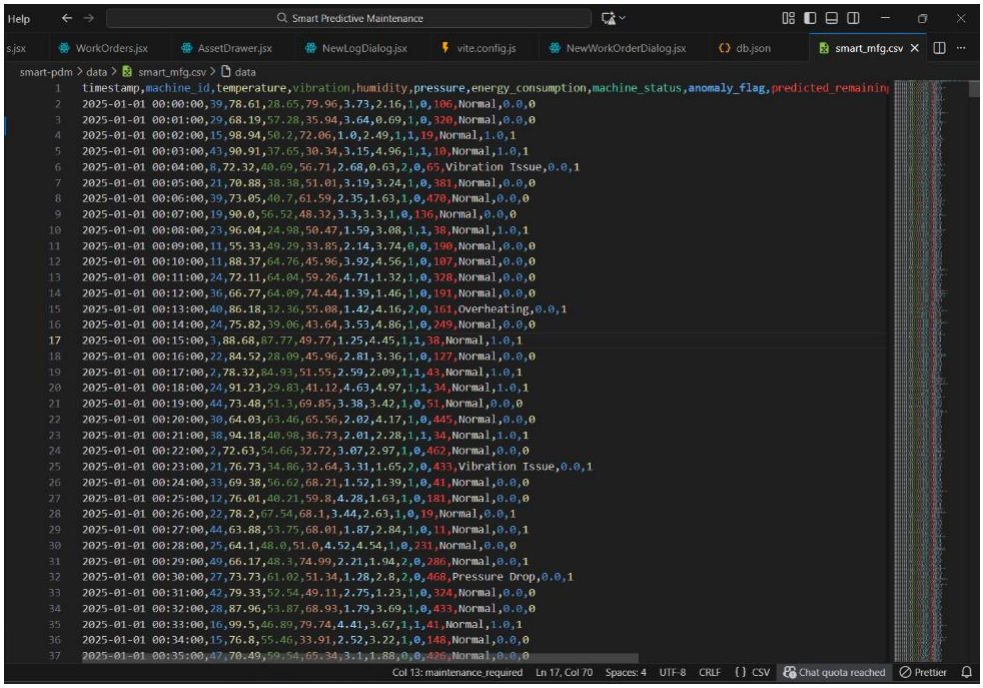
APPENDIX


SMART PREDICTIVE MAINTENANCE
smart-pdm
data
smart_mfg.csv
node_modules
public
logo.png
vite.svg
src
assets
react.svg
auth
AuthContext.jsx
PublicOnly.jsx
RequireAuth.jsx
components
common
Breadcrumbs.jsx
ConfirmDialog.jsx
SectionHeader.jsx
Skeletons.jsx
ToastProvider.jsx
metrics
StatCard.jsx
useCountUp.js
ui
Alert.jsx
badge.jsx
button.jsx
card.jsx
checkbox.jsx
OUTLINE
TIMELINE
0 0

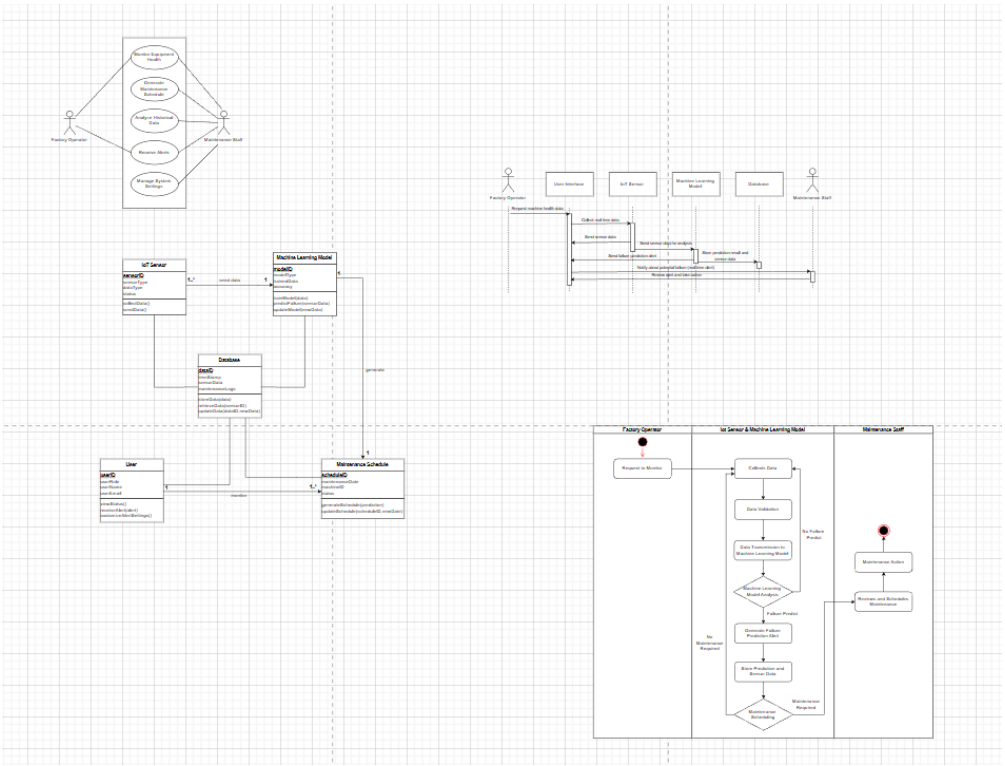

SMART PREDICTIVE MAINTENA...
smart-pdm
src
components
EmptyState.jsx
ErrorBoundary.jsx
GlobalSearch.jsx
NewLogDialog.jsx
NewWorkOrderDialog.jsx
Pagination.jsx
QuickActions.jsx
RecentActivity.jsx
Table.jsx
WorkOrderDrawer.jsx
layouts
ShellLayout.jsx
lib
alerts.js
api.js
query.js
toast.js
utils.js
pages
Alerts.jsx
Assets.jsx
Dashboard.jsx
Login.jsx
NotFound.jsx
Profile.jsx
Settings.jsx
Signup.jsx
OUTLINE
TIMELINE
0 0


SMART PREDICTIVE MAINTENA...
smart-pdm
src
pages
Settings.jsx
Signup.jsx
Telemetry.jsx
WorkOrders.jsx
utils
csv.js
App.css
App.jsx
index.css
main.jsx
MaintenanceLogs.jsx
tools
import-kaggle.js
.env
.gitignore
components.json
db.json
eslint.config.js
index.html
jsconfig.json
package-lock.json
package.json
postcss.config.js
README.md
tailwind.config.js
vite.config.js
OUTLINE
TIMELINE
0 0


Help
Smart Predictive Maintenance
db.json
smart_mfg.csv
Telemetry.jsx
package.json
api.js
StatusBadge.jsx
MaintenanceLogs.jsx
Table.jsx
smart-pdm > src > pages > Telemetry.jsx > ...
export default function Telemetry() {
useEffect(() => {
async function load() {
type: "error",
title: "Failed to load telemetry",
description: err?.message || "API error",
});
}
}
load();
}, [pushToast]);
// Global stats (all rows)
const total = rows.length;
const failures = rows.filter((r) => r.failure === 1).length;
const failureRate = total ? (failures / total) * 100 : 0;
// Filtered data based on selected machine
useEffect(() => {
if (selectedMachine) {
const filteredMachinesData = rows.filter((row) => row.machineId === selectedMachine);
setFilteredData(filteredMachinesData);
setShowGraph(true); // Show graph once a machine is selected
} else {
setShowGraph(false); // Hide graph if no machine is selected
}
}, [selectedMachine, rows]);
/* ------------------------ FILTERED DATA ------------------------ */
const filteredRows = useMemo(() => {
return filteredData.filter((r) => {
// Status filter
if (statusFilter === "normal" && r.failure === 1) return false;
if (statusFilter === "failure" && r.failure !== 1) return false;
// Text search (machine id or timestamp)
if (!query.trim()) return true;
Ln 376, Col 1
Spaces: 2
UTF-8
CRLF
JavaScript JSX
Chat quota reached
Prettier


Help
Smart Predictive Maintenance
s.jsx
WorkOrders.jsx
AssetDrawer.jsx
NewLogDialog.jsx
vite.config.js
NewWorkOrderDialog.jsx
db.json
smart_mfg.csv
smart-pdm > data > smart_mfg.csv > data
timestamp,machine_id,temperature,vibration,humidity,pressure,energy_consumption,machine_status,anomaly_flag,predicted_remaining
2025-01-01 00:00:00,39,78.61,28.65,79.96,3.73,2.16,1,0,106,Normal,0.0,0
2025-01-01 00:01:00,29,68.19,57.28,35.94,3.64,0.69,1,0,320,Normal,0.0,0
2025-01-01 00:02:00,15,98.94,50.2,72.06,1.0,2.49,1,1,19,Normal,1.0,1
2025-01-01 00:03:00,43,90.91,37.65,30.34,3.15,4.96,1,1,10,Normal,1.0,1
2025-01-01 00:04:00,8,72.32,40.69,56.71,2.68,0.63,2,0,65,Vibration Issue,0.0,1
2025-01-01 00:05:00,21,70.88,38.38,51.01,3.19,3.24,1,0,381,Normal,0.0,0
2025-01-01 00:06:00,39,73.05,40.7,61.59,2.35,1.63,1,0,470,Normal,0.0,0
2025-01-01 00:07:00,19,90.0,56.52,48.32,3.3,3.3,1,0,136,Normal,0.0,0
2025-01-01 00:08:00,23,96.04,24.98,50.47,1.59,3.08,1,1,38,Normal,1.0,1
2025-01-01 00:09:00,11,55.33,49.29,33.85,2.14,3.74,0,0,190,Normal,0.0,0
2025-01-01 00:10:00,11,88.37,64.76,45.96,3.92,4.56,1,0,107,Normal,0.0,0
2025-01-01 00:11:00,24,72.11,64.04,59.26,4.71,1.32,1,0,328,Normal,0.0,0
2025-01-01 00:12:00,36,66.77,64.09,74.44,1.39,1.46,1,0,191,Normal,0.0,0
2025-01-01 00:13:00,40,86.18,32.36,55.08,1.42,4.16,2,0,161,Overheating,0.0,1
2025-01-01 00:14:00,24,75.82,39.06,43.64,3.53,4.86,1,0,249,Normal,0.0,0
2025-01-01 00:15:00,3,88.68,87.77,49.77,1.25,4.45,1,1,38,Normal,1.0,1
2025-01-01 00:16:00,22,84.52,28.09,45.96,2.81,3.36,1,0,127,Normal,0.0,0
2025-01-01 00:17:00,2,78.32,84.93,51.55,2.59,2.09,1,1,43,Normal,1.0,1
2025-01-01 00:18:00,24,91.23,29.83,41.12,4.63,4.97,1,1,34,Normal,1.0,1
2025-01-01 00:19:00,44,73.48,51.3,69.85,3.38,3.42,1,0,51,Normal,0.0,0
2025-01-01 00:20:00,30,64.03,63.46,65.56,2.02,4.17,1,0,445,Normal,0.0,0
2025-01-01 00:21:00,38,94.18,40.98,36.73,2.01,2.28,1,1,34,Normal,1.0,1
2025-01-01 00:22:00,2,72.63,54.66,32.72,3.07,2.97,1,0,462,Normal,0.0,0
2025-01-01 00:23:00,21,76.73,34.86,32.64,3.31,1.65,2,0,433,Vibration Issue,0.0,1
2025-01-01 00:24:00,33,69.38,56.62,68.21,1.52,1.39,1,0,41,Normal,0.0,0
2025-01-01 00:25:00,12,76.01,40.21,59.8,4.28,1.63,1,0,181,Normal,0.0,0
2025-01-01 00:26:00,22,78.2,67.54,68.1,3.44,2.63,1,0,19,Normal,0.0,1
2025-01-01 00:27:00,44,63.88,53.75,68.01,1.87,2.84,1,0,11,Normal,0.0,1
2025-01-01 00:28:00,25,64.1,48.0,51.0,4.52,4.54,1,0,231,Normal,0.0,0
2025-01-01 00:29:00,49,66.17,48.3,74.99,2.21,1.94,2,0,286,Normal,0.0,1
2025-01-01 00:30:00,27,73.73,61.02,51.34,1.28,2.8,2,0,468,Pressure Drop,0.0,1
2025-01-01 00:31:00,42,79.33,52.54,49.11,2.75,1.23,1,0,324,Normal,0.0,0
2025-01-01 00:32:00,28,87.96,53.87,68.93,1.79,3.69,1,0,433,Normal,0.0,0
2025-01-01 00:33:00,16,99.5,46.89,79.74,4.41,3.67,1,1,41,Normal,1.0,1
2025-01-01 00:34:00,15,76.8,55.46,33.91,2.52,3.22,1,0,148,Normal,0.0,0
2025-01-01 00:35:00,47,70.49,59.54,65.34,3.1,1.88,0,0,426,Normal,0.0,0
Col 13: maintenance_required
Ln 17, Col 70
Spaces: 4
UTF-8
CRLF
CSV
Chat quota reached
Prettier


Predicta Core
Dashboard
Assets
Work Orders
Logs
Alerts
Telemetry
Settings
Machine 40
Select Machine
Machine 39
Machine 29
Machine 15
Machine 43
Machine 8
Machine 21
Machine 19
Machine 23
Machine 11
Machine 24
Machine 36
Machine 40
Machine 3
Machine 22
Machine 2
Machine 44
Machine 30
Machine 38
Machine 33
2025-01-30 07:01:00
humidity : 37.91
temperature : 76.38
vibration : 32.7
2025-01-10 03:52:00 2025-01-19 17:25:00 2025-01-29 08:45:00 2025-02-07 14:50:00 2025-02-16 09:04:00 2025-02-25 16:16:00 2025-03-11 10:22:00
Search:
Machine ID or timestamp...
Status:
All
MACHINE
TEMP
VIBRATION
HUMIDITY
PRESSURE
POWER
MODE
MAINT.
STATUS
40 86.18 32.36 55.08 1.42 4.16 Overheating Required Normal
2025-01-01 01:03:00 40 51.81 42.04 58.43 3.12 2.28 Normal – Normal
2025-01-01 01:23:00 40 57.59 39.04 58.14 4 2.82 Normal – Normal
2025-01-01 02:09:00 40 68.76 77.81 68.88 1.74 4.09 Normal – Normal
2025-01-01 04:20:00 40 61.37 46.58 45.71 3.41 1.44 Normal – Normal
2025-01-01 04:23:00 40 78.92 59.18 39.32 1.19 3.87 Normal – Normal


Factory Operator
Maintenance Staff
IoT Sensor
Machine Learning Model
Database
User
Maintenance Schedule
Collect Data
Data Validation
Maintenance Action
Maintenance Required

```
import React from "react";
import { BrowserRouter, Routes, Route } from "react-router-dom";

import ShellLayout from "./layouts/ShellLayout.jsx";
import Dashboard from "./pages/Dashboard.jsx";
import Assets from "./pages/Assets.jsx";
import WorkOrders from "./pages/WorkOrders.jsx";
import MaintenanceLogs from "./MaintenanceLogs.jsx";
import Alerts from "./pages/Alerts.jsx";
import Settings from "./pages/Settings.jsx";
import Profile from "./pages/Profile.jsx";
import Telemetry from "./pages/Telemetry.jsx";

import Login from "./pages/Login.jsx";
import Signup from "./pages/Signup.jsx";

import RequireAuth from "./auth/RequireAuth.jsx";
import PublicOnly from "./auth/PublicOnly.jsx";
import ErrorBoundary from "./components/ErrorBoundary.jsx";

export default function App() {
  return (
    <BrowserRouter>
      <ErrorBoundary>
        <Routes>
          {/* Public routes */}
          <Route
            path="/login"
            element={
              <PublicOnly>
                <Login />
              </PublicOnly>
            }
          />
          <Route
            path="/signup"
            element={
              <PublicOnly>
                <Signup />
              </PublicOnly>
            }
          />

          {/* Protected app routes */}
          <Route
            element={
              <RequireAuth>
                <ShellLayout />
              </RequireAuth>
            }
          >
            <Route index element={<Dashboard />} />
            <Route path="assets" element={<Assets />} />
            <Route path="work-orders" element={<WorkOrders />} />
            <Route path="logs" element={<MaintenanceLogs />} />
            <Route path="alerts" element={<Alerts />} />
```

```
            <Route path="telemetry" element={<Telemetry />} />
            <Route path="settings" element={<Settings />} />
            <Route path="profile" element={<Profile />} />
          </Route>

          {/* Fallback */}
          <Route path="*" element={<Login />} />
        </Routes>
      </ErrorBoundary>
    </BrowserRouter>
  );
}
```